PHYSICS

1) The potential energy of a particle in a force field is $U = \frac{A}{r^3} - \frac{B}{r^2}$, where A and B are positive constants and r is the distance of particle from the centre of field. For the equilibrium, the distance of particle is :-

(1) $\frac{2A}{3B}$

(2) $\frac{3A}{2B}$

(3) $\frac{A}{3B}$

(4) $\frac{A}{2B}$

2) The rate of mass of the gas emitted from rear of a rocket is initially 0.1 kg/sec. If the speed of the gas relative to the rocket is 50 m/sec and mass of the rocket is 2 kg, then the acceleration of the rocket (in m/sec^2) is :- (Rocket is in gravity free space)

(1) 5

(2) 5.2

(3) 2.5

(4) 25

@EXAMLIX

3) Ratio of weight of a man in a stationary lift and weight of a man in a lift moving downward with an acceleration is 3 : 2 then acceleration of lift is :

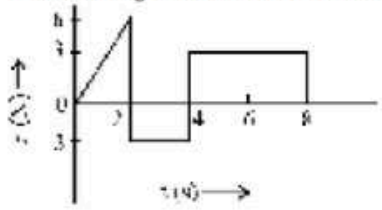
(1) g / 3

(2) g / 2

(3) g

(4) 2g

4) The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-



(1) 24 Ns

(2) 20 Ns

(3) 12 Ns

(4) 6 Ns

5) A block of metal weighing 2 kg is resting on a frictionless horizontal plane. It is struck by a jet releasing water horizontally at a rate of 1 kgs^{-1} and a speed of 5 ms^{-1} horizontally. The initial acceleration of the block will be :- (Assume jet comes to rest after striking the block)

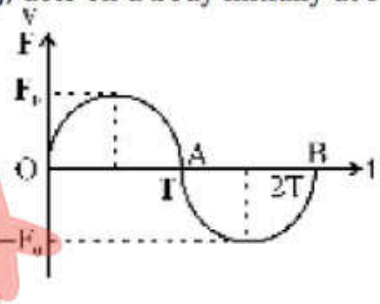
(1) 2.5 ms^{-2}

(2) 5 ms^{-2}

(3) 10 ms^{-2}

(4) 20 ms^{-2}

6) A unidirectional force F varying with time t as shown in the fig, acts on a body initially at rest, for a short duration 2T. Then the velocity acquired by the body is :-



(1) $\frac{\pi F_0 T}{4m}$

(2) $\frac{\pi F_0 T}{2m}$

(3) $\frac{F_0 T}{4m}$

(4) zero

@EXAMLIX

7) A monkey of mass 40 kg climbs on a massless rope of breaking strength 600 N. The rope will break if the monkey :-

(1) Climbs up with a uniform speed of 5 m/s

(2) Climbs up with an acceleration of 5 m/s^2

(3) Climbs down with an acceleration of 4 m/s^2

(4) Climbs down with a uniform speed of 5 m/s

8) An engine pumps up 100 kg of water through a height of 10 m in 5 s. If the efficiency of the engine is 60%, the power of the engine is (g = 10 m s^{-2})

(1) 3.33 kW

(2) 0.33 kW

(3) 2.00 kW

(4) 5.00 kW

9) A 70 kg. man jumps to a height of 0.8 m. Calculate impulse provided by ground to man.

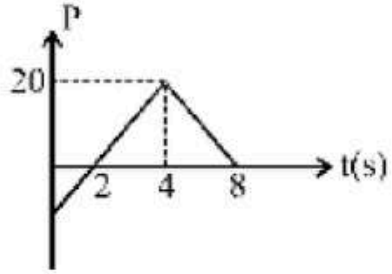
(1) 384.6 kg m/sec

(2) 277.2 kg m/sec

(3) 237.4 kgm/sec

(4) 309.6 kg m/sec

10) From the following momentum-time graph, find the magnitude of negative force.



(1) 5 N

(2) 10 N

(3) 4 N

(4) 2 N

11) A body of mass 1 kg rests on a horizontal floor with which it has a coefficient of static friction $\frac{1}{\sqrt{3}}$. It is desired to make the body move by applying the minimum possible force F Newton. The value of F will be _____.(Round off to the Nearest Integer) [Take g = 10 ms^{-2}]

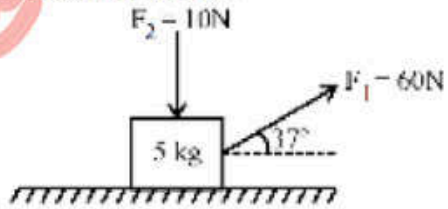
(1) 5

(2) 6

(3) 4

(4) 1

12) If (μ = 0.5) then find out acceleration of block :



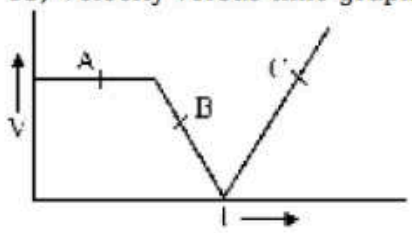
(1) 25 m/s^2

(2) 12 m/s^2

(3) 7.2 m/s^2

(4) 0

13) Velocity versus time graph of a body is shown in Fig. It explains that :-



@EXAMLEX

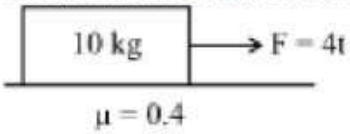
(1) At B force is zero

(2) At B there is a force but towards motion

(3) At B there is a force but opposes the motion

(4) Forces are equal at A, B and C

14) A block of mass 10 kg is kept over a rough surface and a force F = (4t) N is applied on it. Then the minimum value of t for which the block will start moving is ($g = 10\ m/s^2$)



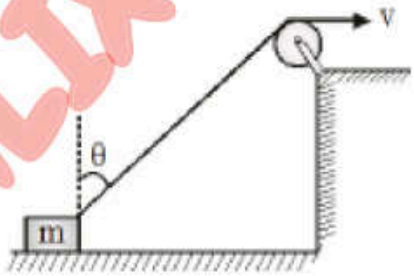
(1) 4 sec.

(2) 10 sec.

(3) 8 sec.

(4) 5 sec.

15) A block is dragged on a smooth plane with the help of a rope which moves with a velocity v as shown in figure. The horizontal velocity of the block is:



(1) v

(2) $v \sin\theta$

(3) $\frac{v}{\sin\theta}$

(4) $\frac{v}{\cos\theta}$

16) A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0. The instantaneous velocity of the car is proportional to

(1) $t^{1/2}$

(2) $t^{-1/2}$

(3) t^2

(4) t

17) A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})\ ms^{-1}$ under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})\ N$. The instantaneous power applied to the particle is:-

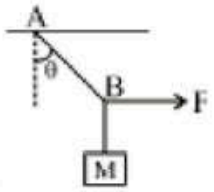
(1) 200 W

(2) 320 W

(3) 140 W

@EXAMLIX

(4) 170 W



18) If the system is in equilibrium then tension in the string AB is

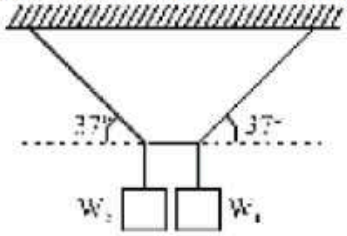
(1) F sinθ

(2) F/sinθ

(3) F cos θ

(4) $\frac{F}{\cos\theta}$

19) In a given figure, system is in equilibrium. If W_1 = 300 N. Then W_2 is equal to :-



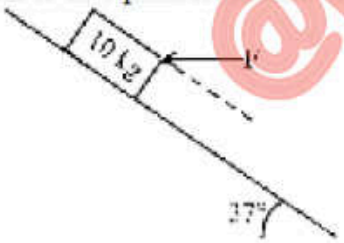
(1) 500 N

(2) 400 N

(3) 670 N

(4) 300 N

20) If system is in equilibrium find external force F :-



(1) 80 N

(2) 60 N

(3) 75 N

(4) 100 N

21) A block of mass 4 kg is suspended through two light spring balances A and B. Then A and B will

@EXAMLIX



read respectively :-

(1) 4 kg and zero kg

(2) Zero kg and 4 kg

(3) 4 kg and 4 kg

(4) 2 kg and 2 kg

22) An iron ball of mass 100 g moving with a velocity of 10 m/s collides with a wall at an angle 30° with wall & rebounds at the same angle with the same speed. If period of contact between the ball & the wall is 0.1 sec. the force experienced by the wall is

(1) 10 N

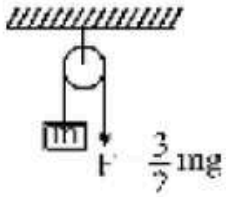
(2) 100 N

(3) 1 N

(4) 0.1 N

@EXAMLIX

23) In the arrangement shown acceleration of the block is



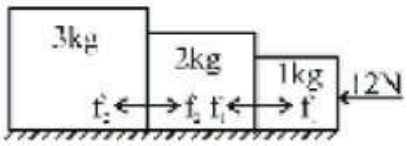
(1) Zero

(2) g/2

(3) g

(4) 2g

24) Contact forces f_1 and f_2 are :-



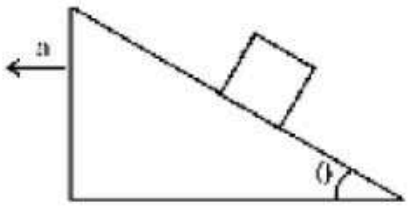
(1) 10N, 6N

(2) 6N, 10N

(3) 3N, 2N

(4) 2N, 3N

25) A block of mass m is resting on a wedge of angle θ as shown in the figure. With what minimum acceleration a should the wedge move so that the mass m falls freely :-



(1) g

(2) g cos θ

(3) g cot θ

(4) g tan θ

26) Column I represents work done by forces and column II represents change in kinetic energy Δk, change in potential energy ΔU, change in mechanical energy ΔE. Then match the two column.

Column I		**Column II**	
A.	Work done by conservative force	**P.**	ΔK
B.	Work done by non-conservative force	**Q.**	-ΔU
C.	Work done by all forces	**R.**	ΔE

(1) A-Q, B-R, C-P

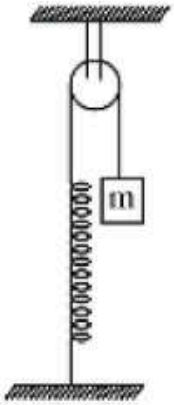
(2) A-R, B-P, C-Q

(3) A-P, B-R, C-Q

(4) A-P, B-Q, C-R

@EXAMLIX

27) Consider the situation shown in figure initially the spring is unstreched when the system is released from rest. Assuming no friction in pulley, find maximum elongation of spring?



(1) $\frac{mg}{K}$

(2) $\frac{mg}{2K}$

(3) $\frac{2mg}{K}$

(4) $\frac{4mg}{K}$

28) A particle moves on a rough horizontal ground with some initial velocity say v_0. If 3/4 of its kinetic energy is lost in friction in time t_0 then coefficient of friction between the particle and the ground is

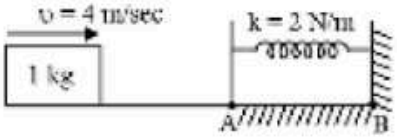
(1) $\frac{v_0}{2gt_0}$

(2) $\frac{v_0}{4gt_0}$

(3) $\frac{3v_0}{4gt_0}$

(4) $\frac{v_0}{gt_0}$

29) A 1 kg block is moving on smooth track with speed υ it strikes with spring of spring constant k = 2 N/m and surface below spring is rough with μ = 0.2 then maximum compression in spring will be :



(1) 1 m

(2) 2 m

(3) $2\sqrt{2}$ m

(4) none of these

30) A force $\vec{F} = 3\hat{i} + c\hat{j} + 2\hat{k}$ acting on a particle causes a displacement $\vec{d} = -4\hat{i} + 2\hat{j} - 3\hat{k}$. If the work done is 6J, then the value of c will be :-

(1) 12

(2) 6

(3) 1

(4) 0

31) The figure shows a smooth curved track terminating in a smooth horizontal part. A spring of spring constant 400 N/m is attached at one end to a wedge fixed rigidly with the horizontal part. A 40 g mass is released from rest at height of 5 m on the curved track. The maximum compression of the spring will be



(1) 1 cm

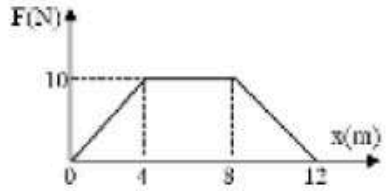
(2) 2 cm

@EXAMLIX

(3) 5 cm

(4) 10 cm

32) A particle of mass 0.1 kg is subjected to a force which varies with distance as shown in the figure. If it starts its journey from rest at x = 0, its velocity at x = 12 m is :-



(1) 0 m/s

(2) $20\sqrt{2}$ m/s

(3) $20\sqrt{3}$ m/s

(4) 40 m/s

33) A body of mass M is dropped from a height h on a sand floor. If the body penetrates x cm into the sand, the average resistance offered by the sand to the body is :

(1) $Mg\left(\frac{h}{x}\right)$

(2) $Mg\left(1+\frac{h}{x}\right)$

(3) Mgh + Mgx

(4) $Mg\left(1-\frac{h}{x}\right)$

@EXAMLIX

34) In stretching a spring by 2 cm energy stored is given by U, then stretching by 10 cm energy stored will be :

(1) U

(2) 5U

(3) U/25

(4) 25U

35) A uniform chain of length L and mass M is lying on a smooth table and $\frac{1}{3}$ of it's length is hanging down over the edge of the table. If g is the acceleration due to gravity the workdone to pull the hanging part on the table is :-

(1) $\frac{MgL}{3}$

(2) $\frac{MgL}{6}$

(3) $\frac{2MgL}{9}$

(4) $\frac{MgL}{18}$

36) If momentum is increased by 100%, the kinetic energy will increase by :-

(1) 100%
(2) 200%
(3) 300%
(4) 400%

37) A stone is projected vertically so that it reaches to maximum height h. The ratio of its kinetic and potential energy at height 4h/5 is :–

(1) 5/4
(2) 4/5
(3) 1/4
(4) 4/1

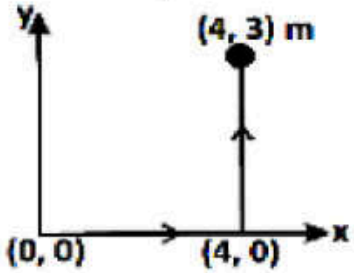
38) A string is used to pull a block of mass m vertically up by a distance h at a constant acceleration



g/2. The work done by tension in the string is :

(1) $\frac{+3mgh}{2}$
(2) $\frac{-mgh}{4}$
(3) $+\frac{5}{4}mgh$
(4) +mgh.

39) A force $\vec{F} = \left(3\hat{i} + 4\hat{j}\right)$ N acts on a particle moving in x-y plane. Starting from origin the particle first goes along x-axis to the point (4,0)m and then parallel to the y-axis to the point (4,3)m. The total work done by the force on the particle is :



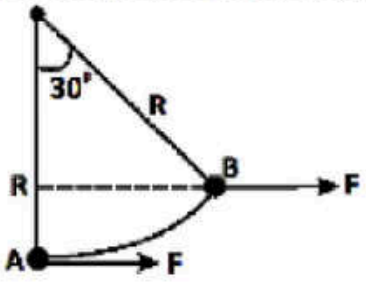
(1) + 12 J
(2) - 6 J

@EXAMLIX

(3) + 24 J

(4) –12 J.

40) A block of mass m is pulled along a circular arc, by means of a constant horizontal force F as shown. Work done by this force in pulling block from A to B is :



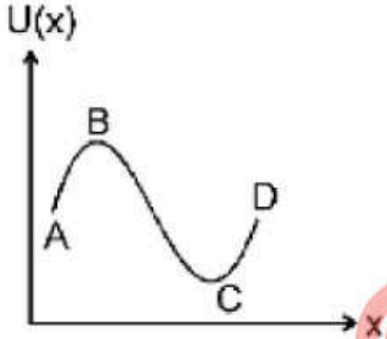
(1) $\frac{FR}{2}$

(2) FR

(3) $\frac{\sqrt{3}}{2}FR$

(4) mgR.

41) The potential energy of a particle varies with distance x as shown in the graph.



The force acting on the particle is zero at

(1) C

(2) B

(3) B and C

(4) A and D

42) The potential energy of a particle varies with x according to the relation $U(x) = x^2 - 4x$. The point x = 2 is a point of :

(1) stable equilibrium

(2) unstable equilibrium

(3) neutral equilibrium

(4) none of above

43) A block is released from rest from top of a rough curved track as shown in figure. It comes to rest at some point on the horizontal part. If its mass is 200 gm, calculate work by friction in joules.

@EXAMLIX



(1) +2 J

(2) -2 J

(3) +4 J

(4) -4 J

44) Two springs have their force constant as k_1 and $k_2(k_1 > k_2)$. When they are stretched by the same force up to equilibrium -

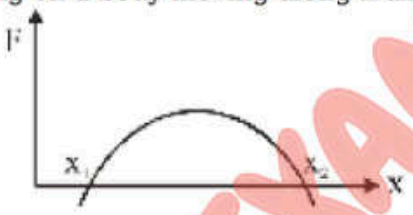
(1) No work is done by this force in case of both the springs

(2) Equal work is done by this force in case of both the springs

(3) More work is done by this force in case of second spring

(4) More work is done by this force in case of first spring

45) The force acting on a body moving along x-axis varies with the position of the particle as shown in the figure.



The body is in stable equilibrium at :-

(1) $x = x_1$

(2) $x = x_2$

(3) both x_1 and x_2

(4) neither x_1 nor x_2

@EXAMLIX

CHEMISTRY

1) Hybridization of XeO_3 and XeO_2F_2 is :-

(1) sp^3 and sp^3d

(2) sp^2 and sp

(3) sp^3d^2 and sp^3d^3

(4) None

2) Which of the following has $p_\pi - d_\pi$ bonding

(1) NO_3^-

(2) SO_3^{-2}

(3) BO_3^{-3}

(4) CO_3^{-2}

3) Correct statement regarding this reaction :
$BF_3 + NH_3 \rightarrow [F_3B \leftarrow NH_3]$

(1) Hybridisation of only N changes
(2) Hybridisation of only B changes
(3) Hybridisation of N and B both change
(4) None of these

4) Which of the following species are hypervalent?
(I) ClO_4^- (II) BF_3 (III) SO_4^{2-} (IV) CO_3^{2-}

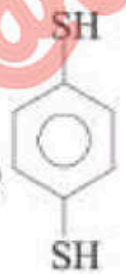
(1) I, II, III
(2) I, III
(3) III, IV
(4) I, III, IV

5) For which of the following molecule significant $\mu \neq 0$?

(a) 

(b) 

(c) 

(d) 

@EXAMLIX

(1) (a) and (b)
(2) only (c)
(3) (c) and (d)
(4) only (a)

6) In HI molecule if observed dipole moment in 1.2 Debye and also the interionic distance between atoms (d_{HI}) is 1A°. The percentage covalent character in HI bond will be :

(1) 25%
(2) 75%
(3) 50%
(4) 12.5%

7) Given below are two statements

Statement-I : The boiling points of the following hydrides of group 16 elements increases in the order-

$H_2O < H_2S < H_2Se < H_2Te$

Statement-II : The boiling points of these hydrides increase with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

8) The geometrical arrangement of orbitals and shape of I_3^- are respectively:

(1) Trigonal bipyramidal geometry, linear shape

(2) Hexagonal geometry, T-shape

(3) Triangular planar geometry, triangular shape

(4) Tetrahedral geometry, pyramidal shape

9) BF_3 and NF_3 both are covalent compounds but NF_3 is polar whereas BF_3 is non-polar. This is because:

(1) Nitrogen atom is smaller than boron atom

(2) N—F bond is more polar than B—F bond

(3) NF_3 is pyramidal whereas BF_3 is trigonal planar

(4) BF_3 is electron deficient whereas NF_3 is not

@EXAMLIX

10) What will be the correct bond length order of Cl–O bond among the following ?

(1) $ClO^- > ClO_2^- > ClO_3^- > ClO_4^-$

(2) $ClO_4^- > ClO_3^- > ClO_2^- > ClO^-$

(3) $ClO^- > ClO_2^- < ClO_3^- > ClO_4^-$

(4) $ClO^- > ClO_2^- > ClO_3^- < ClO_4^-$

11) The boiling point of a compound is raised by –

(1) Intermolecular hydrogen bonding

(2) High volatility

(3) Intramolecular hydrogen bonding

(4) Non-polarity

12) In which of the following species maximum atom lie in same plane?

(1) XeO_2F_2

(2) PCl_5

(3) AsH_4^+

(4) XeF_4

13) Which one of the following contains both ionic and covalent bonds?

(1) C_6H_5Cl

(2) H_2O

(3) NaOH

(4) CO_2

14) Which among the following molecules have sp^3d hybridisation with one lone pair of electrons on the central atom?

(i) SF_4 (ii) $[PCl_4]^+$

(iii) XeO_2F_2 (iv) $ClOF_3$

(1) (i), (ii) and (iii) only

(2) (i), (iii) and (iv) only

(3) (i) and (iii) only

(4) (iii) and (iv) only

15) Which molecular orbital have same energy :-

(1) $\pi 2p_x = \sigma 2s$

(2) $\pi 2p_y = \sigma 2s$

(3) $\pi 2p_x = \pi 2p_y$

(4) $\sigma 2s = \sigma^* 2s$

16) Find the incorrect statement :-

(1) Number of hybrid orbitals produced is equal to the number of atomic orbitals that get hybridised.

(2) Hybrid orbitals are equivalent in energy and shape

(3) Hybrid orbitals are more effective in forming sigma bonds than pure atomic orbitals

(4) In hybridisation only half filled atomic orbitals participates

17) Identify the molecules in which polar as well as non polar bonds are present :-

(A) H_2O_2 (B) N_2H_4 (C) $H_2S_2O_6$

(D) PH_3 (E) NH_4Cl (F) HI

(G) O_2F_2

(1) A, B, C, E, F

(2) A, B, G only

(3) A, B, C

(4) A, B, C, G

18) Order of bond angle is :-

@EXAMLIX

(a) OF_2 (b) Br_2O (c) Cl_2O

(1) a < c < b

(2) a < b < c

(3) c < b < a

(4) All are equal

19)

In the following which of the two are paramagnetic
(a) N_2 (b) CO (c) B_2 (d) NO_2
Correct answer is :-

(1) a and c

(2) b and c

(3) c and d

(4) b and d

20) Which of the following two are isostructural :-

(1) XeF_2, IF_2^-

(2) ClF_3, CO_2

(3) CO_3^{2-}, SO_3^{2-}

(4) PCl_5, ICl_5

@EXAMLIX

21) Which statements is/are incorrect among the following in MOT?
(I) The combining atomic orbitals must have same (or) nearly same energy.
(II) The combining molecular orbitals must have the same symmetry about the molecular axis.
(III) Both sigma and pi molecular orbitals are symmetrical around the bond axis.
(IV) Sigma molecular orbitals symmetrical around the bond axis and pi molecular orbitals are not symmetrical.

(1) II, III only

(2) II, IV only

(3) I, II and IV only

(4) Only III

22) The hybridisation of N in NH_3, NH_4^+ and NO_2^+ is –

(1) sp, sp^3, sp^3

(2) sp^3, sp, sp^2

(3) sp, sp^2, sp^3

(4) sp^3, sp^3, sp

23) **Assertion (A):** LiI is more covalent than LiF.
Reason (R): Polarizability of anion increases from F^- to I^- due to increasing size.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true but R is not correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

24) Select correct order :-

(1) LiF < NaF < KF < RbF (Lattice energy)

(2) NaI > NaBr > NaCl > NaF (Ionic character)

(3) $K^+ < Ca^{2+} < Cd^{2+}$ (Polarising power)

(4) $S^{2-} < O^{2-} < F^-$ (Polarizability)

25) Which of the following is correct order of extent of H-bonding (No. of H-Bond)

(1) $HF > H_2O > H_2O_2$

(2) $H_2O_2 > H_2O > HF$

(3) $H_2O > H_2O_2 > HF$

(4) All are incorrect

26)

Which one of the following has maximum lattice energy?

(1) A□F_3

(2) MgF_2

(3) NaF

(4) NaC□

27) Phosphorus pentachloride in the solid exists as :-

(1) PCl_5

(2) PCl_4^+, Cl^-

(3) PCl_4^+, PCl_6^-

(4) $PCl_5 \cdot Cl_2$

28) Select the incorrect match :-

(1) O-nitrophenol > P-nitrophenol ⇒ Vapour pressure

(2) O-nitrophenol < P-nitrophenol ⇒ Boiling point

(3) $CH_2–OH$ $CH_2–OH$ < $CH_2–OH$ $CH–OH$ $CH_2–OH$ ⇒ Viscosity

(4) $H_2O < HF$ ⇒ Boiling point

@EXAMLIX

29) Which of the following is the strongest attraction between molecules / ions :

(1) H_2 — H_2

(2) H–Cl – C_6H_6

(3) H–Cl – H–Cl

(4) Na – Cl–H

30) Correct order of covalent character :-

(1) $BeCl_2 > MgCl_2 < CaCl_2 < SrCl_2$

(2) $BeCl_2 < CaCl_2 < SrCl_2 < MgCl_2$

(3) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2$

(4) $SrCl_2 > BeCl_2 > CaCl_2 > MgCl_2$

31) Which of following has covalent and coordinate bonds both ?
(a) $N_2H_5^+$ (b) H_2CO_3 (c) N_2O_5
(d) H_3O^+ (e) SiF_6^{2-}
Select the correct answer :

(1) a, d only

(2) a, c, d, e only

(3) a, c, d only

(4) a, b, d only

32) In O_2^-, O_2 and O_2^{2-}, the total number of antibonding electrons respectively are

(1) 7, 6, 8

(2) 1, 0, 2

(3) 6, 6, 6

(4) 8, 6, 8

33) Which of the following diatomic molecular species has only π bonds according to Molecular Orbital Theory ?

(1) O_2

(2) N_2

(3) C_2

(4) Be_2

34) The correct order of increasing s-character (in percentage) in the hybrid orbitals of following molecule/ion is :-
(I) CO_3^{-2} (II) XeF_4 (III) I_3^-

@EXAMLIX

(IV) NCl_3 (V) $BeCl_2(g)$

(1) II < III < IV < I < V

(2) II < IV < III < V < I

(3) III < II < I < V < IV

(4) II < IV < III < I < V

35) **Statement-I :-** Formal charge help in the selection of the lowest energy structure from a number of possible Lewis structures for a given species.
Statement-II :- Generally the lowest energy structure is the one with the smallest formal charge on the atoms.

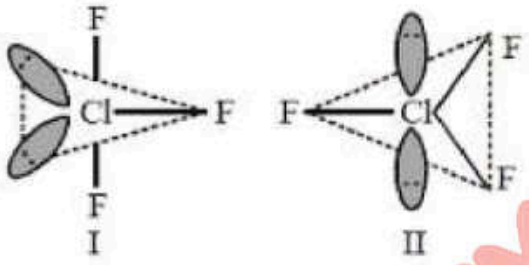
(1) Both Statement-I and Statement-II are incorrect.

(2) Both Statement-I and Statement-II are correct.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

36) Which of the following structure is most stable ?



Choose the correct option.

(1) Only I

(2) Only II

(3) Only III

(4) All three have same stability

37) Ionic conductances of hydrated M+ ions are in the order –

(1) Li^+ (aq)>Na^+ (aq)>K^+ (aq)>Rb^+(aq) > Cs^+ (aq)

(2) Li^+(aq)>Na^+(aq)<K^+(aq) < Rb^+(aq) < Cs^+(aq)

(3) Li^+(aq)>Na^+(aq)> K^+(aq)>Rb^+(aq) < Cs^+ (aq)

(4) Li^+ (aq)<Na^+ (aq)<K^+ (aq)<Rb^+ (aq)<Cs^+ (aq)

38) **Statement 1 (S_1) :** CO_2 and SO_2 both are polar compounds.
Statement 2 (S_2) : Both have polar bonds.

(1) S_1 & S_2 both are correct

(2) S_1 and S_2 both are incorrect

(3) S_1 is correct but S_2 is incorrect

(4) S_1 is incorrect but S_2 is correct

@EXAMLIX

39) If all the fluorine atoms of XeF_4 are replaced by same number of oxygen atoms then which of the following property does not change in above process:

(1) Oxidation state of Xe
(2) Hybridisation of Xe
(3) Planarity of molecule
(4) Dipole moment of molecule

40) Which of the following pair have same hybridisation ?

(1) ClO_4^- & ClO_2^-
(2) SF_4 & CCl_4
(3) BF_3 & NF_3
(4) CO_2 & SO_2

41) Select correct order of bond strength by overlapping of atomic orbital :

(1) 1s – 1s > 1s – 2s > 1s – 2p
(2) 2s – 2s > 2s – 2p > 2p – 2p
(3) 2s – 2p > 2s – 2s > 2p – 2p
(4) 1s – 1s > 1s – 2p > 1s – 2s

42) An ionic compound $A^+ B^-$ is most likely to be formed when -

(1) Ionization energy of A is low
(2) Electron affinity of B is high
(3) Electron affinity of B is low
(4) Both (1) and (2)

@EXAMLIX

43)

Which one is polar covalent molecule ?

(1) SO_3
(2) NaCl
(3) H_2O
(4) F_2

44) Which of the following pairs of species have unpaired electrons in antibonding molecular orbitals:

(1) C_2, O_2
(2) O_2, He_2^+
(3) He_2^+, C_2
(4) N_2, O_2^-

45) In which of the following compound all the bond angles are equal :-

(1) SF_4

(2) CCl_4

(3) $CHCl_3$

(4) XeF_6

BOTANY

1) In fungi which of the following is not produced during sexual reproduction ?

(1) Oospore

(2) Zoospore

(3) Ascospore

(4) Basidiospore



2) Which of the following statement is correct regarding given diagram ?

(1) Two plant cells are fusing

(2) Two bacterial cells are fusing to form zygote

(3) Binary fission in bacterial cell

(4) All of the above

3) Identify the figures with features and select the correct option :-



(a)



(b)



(c)

(i) The mycelium is aseptate

(ii) Asexual spores are generally not found

(iii) Sexual spores are produced endogenously.

Options :-

(1) a–i, b–ii, c–iii

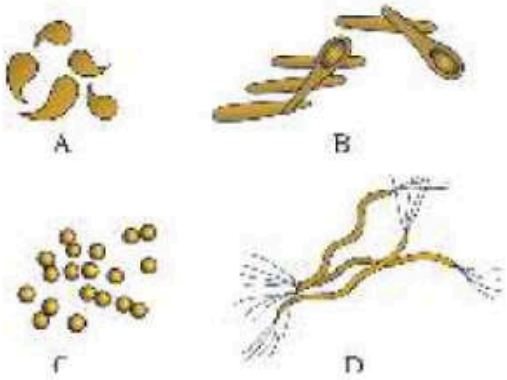
(2) a–i, b–iii, c–ii

(3) a–ii, b–i, c–iii

(4) a–iii, b–ii, c–i

4) Identify bacteria with given figure according to their shape :-

@EXAMLIX



	A	B	C	D
(1)	Bacilli	Cocci	Spirilla	Vibrio
(2)	Cocci	Spirilla	Vibrio	Bacilli
(3)	Spirilla	Vibrio	Bacilli	Cocci
(4)	Vibrio	Bacilli	Cocci	Spirilla

(1) 1

(2) 2

(3) 3

(4) 4

5) Find the correct match

(1) Sex organs absent - Ascomycetes

(2) Spore dispersal in slime moulds - by air currents

(3) Coprophilous Fungi - Grow on wood

(4) Aggregation of slime moulds - Mycelium

@EXAMLIX

6) Which one one of the following matches is correct ?

(1)	*Alternaria*	*Sexual reproduction absent*	*Deuteromycetes*
(2)	*Mucor*	Reproduction by Conjugation	Ascomycetes
(3)	*Agaricus*	Parasitic fungus	Basidiomycetes
(4)	*Phytophthora*	Aseptate mycelium	Basidiomycetes

(1) 1

(2) 2

(3) 3

(4) 4

7) Consider the table given below and answer what are A, B and C:

Characters	Monera	Protista	Fungi	Plantae
Cell wall	Peptidoglycan	Present in some	A	Cellulose
Nuclear Membrane	B	Present	Present	Present
Body organisation	Cellular	Cellular	C	Tissue/ Organ

(1) A: Peptidoglycan; B: Present; C: Cellular

(2) A: Chitin; B: Present; C: Loose Tissue

(3) A: Chitin; B: Absent; C: Loose Tissue

(4) A: Peptidoglycan; B: Absent; C: Cellular

8) Which one of the following is not a criteria of five kingdom system?

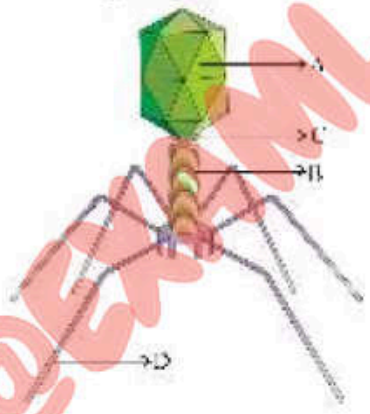
(1) Cell type

(2) Body organisation

(3) Reserve food materials

(4) Phylogeny

9) Examine the figure and select the right option out of 1-4, in which all the four structure, A, B, C and D are identify correctly :



	A	B	C	D
(1)	Head	Sheath	Collar	Tail fibres
(2)	Collar	Head	Sheath	Tail fibres
(3)	Sheath	Collar	Head	Tail fibres
(4)	Head	Collar	Sheath	Tail fibres

(1) 1

(2) 2

(3) 3

(4) 4

10) **Assertion (A)** : The two kingdom classification, used for a long time, was found to be inadequate.
Reason (R) : Two kingdom system of classification did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and green algae and fungi.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

@EXAMLIX

(3) A is true, but R is false.

(4) A is false, but R is true.

11) Choose the incorrect statement :

(1) The three-domain system has also been proposed that divides the kingdom Monera into two domain.

(2) Earlier classification system included bacteria, BGA, fungi, mosses, ferns, Gymnosperm and the angiosperm under Plant.

(3) The body organisation of fungi is multicellular and loose tissue.

(4) The main criteria of five kingdom classification is cell structure, body organisation, mode of nutrition but not Reproduction and Phylogenetic relationship.

12)

Match column-I with column-II and choose the correct combination from the options given below :-

Column-I		**Column-II**	
(A)	Phycomycetes	(i)	Algal fungi
(B)	Basidiomycetes	(ii)	Sac fungi
(C)	Ascomycetes	(iii)	Club fungi
(D)	Deuteromycetes	(iv)	Imperfect fungi

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(i), B-(iii), C-(ii), D-(iv)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(i), B-(ii), C-(iv), D-(iii)

@EXAMLIX

13) Members of which group are related with basidiomycetes ?

(1) Mushroom, *Ustilago, Aspergillus*

(2) *Puccinia, Agaricus, Aspergillus*

(3) *Puccinia, Ustilago, Agaricus*

(4) *Ustilago, Aspergillus, Agaricus*

14) Arrange the following events in correct sequence for sexual cycle of fungi ?
(a) Karyogamy
(b) Meiosis
(c) Zygote formation
(d) Plasmogamy
(e) Dikaryon formation

(1) d, e, a, c, b

(2) d, a, e, c, b

(3) e, a, d, b, c

(4) a, d, e, c, b

15) ***Assertion :-*** In members of basidiomycetes, asexual spores are conidia.
Reason :- Conidia formed endogenously on special mycelium called conidiophores.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

16) How many statements are true for chrysophytes?
(i) They are microscopic and float actively in water current.
(ii) Formation of diatomaceous soil.
(iii) Diatoms are chief producer in terrestial ecosystem.
(iv) Presence of soap box like cell wall.

(1) i, ii, iii, iv
(2) i, ii, iv
(3) ii, iv
(4) iii, iv

17) Which of following statement is incorrect in respect of members of kingdom monera ?

(1) They show the most extensive metabolic diversity
(2) Majority of bacteria are autotrophs
(3) Eubacteria play a great role in recycling of nutrients
(4) BGA have the ability to fix nitrogen

@EXAMLIX

18) Branched, Aseptate coenocytic mycelium present in :-

(1) Aspergillus
(2) Albugo
(3) Penicillium
(4) Erysiphe

19) The subunit of capsid is called :-

(1) capsomere
(2) core
(3) nucleoside
(4) nucleotide

20) Under favourable conditions slime moulds form:-

(1) protonema
(2) *Plasmodium*
(3) mycelium

(4) fruiting bodies

21) A fungus which causes 'white spot disease' in the member of mustured (cruciferae family) is :-

(1) Puccinia
(2) Erysiphae
(3) Albugo
(4) Rhizopus and mucor

22)

Which of the following is a parasitic protozona?

(1) *Paramoecium*
(2) *Gonyaulax*
(3) *Euglena*
(4) *Trypanosoma*

23) Which is not an example of phycomycetes :-

(1) *Albugo*
(2) *Mucor*
(3) Mushroom
(4) *Rhizopus*

24) Which fungi produce exogenous sexual spore?

(1) *Neurospora*
(2) *Aspergillus*
(3) *Agaricus*
(4) *Alternaria*

25) Find out the **incorrect** statement in the following :-

(1) Heterotrophic bacteria are most abundant in nature.
(2) Bacterial structure is very complex and they have very simple behaviour.
(3) Bacteria reproduce mainly by binary fission.
(4) Archaebacteria differ from other bacteria in having different cell wall structure.

26) Kingdom of unicelluler eukaryotes and unicelluler prokaryotes respectively is:-

(1) Monera and protista
(2) Monera and plantae
(3) Protista and monera
(4) Protista and mycota

27) According to five kingdom classification, multicellular eukaryotic producer are placed in?

@EXAMLIX

(1) Three kingdom
(2) Two kingdom
(3) One kingdom
(4) Four kingdom

28) All prokaryotes have a cell wall surrounding the cell membrane except in

(1) Bacteria
(2) Plant
(3) Fungi
(4) Mycoplasma

29) **Statement-1 :-** The three-domain system divides the Kingdom Monera into three domains.
Statement-2 :- Five Kingdom system of Classification proposed by R.H. Whittaker.

(1) Only statement-1 is correct
(2) Only statement-2 is correct
(3) Both statement-1 and statement-2 are correct
(4) Both statement-1 and statement-2 are incorrect

30) **Assertion :** Deuteromycetes are known as imperfect fungi.
Reason : Only the sexual or vegetative phases of these fungi are known.

(1) Both A and R true and R is the correct explanation of A
(2) Both A and R true but R is not correct explanation A
(3) A is true but R is false
(4) A is false and R is true

@EXAMLIX

31) **Statement 1:-** Slime moulds are saprophytic protists.
Statement 2:- During unfavourable conditions, slime moulds form an aggregation called plasmodium.

(1) Only statement 1 is correct
(2) Only statement 2 is correct
(3) Both statement 1 and statement 2 are correct
(4) Both statement 1 and statement 2 are incorrect

32) Match the columns I and II, and choose the correct combination from the options given.

	Column-I		**Column-II**
A.	Chief producer of ocean	I.	Slime mould
B.	Red tides	II.	*Euglena*
C.	Pelicle layer	III.	*Gonyaulax*
D.	Plasmodium	IV.	Diatoms

(1) A- III, B-I, C-II, D-IV
(2) A- I, B-II, C-III, D-IV
(3) A- IV, B-III, C-II, D-I
(4) A- II, B-IV, C-I, D-III

33) Potato spindle tuber diease is cause by :-

(1) Virus
(2) Lichen
(3) Prions
(4) Viroid

34) Cynobacteria are used in agricultural fields for crops improvements because of :

(1) Respiration
(2) Algal blooms
(3) N_2 fixation
(4) All of these

35) In Carl Woese classification, how many kingdom have prokaryotic organisms ?

(1) Two kingdom
(2) One kingdom
(3) Three kingdom
(4) Prokaryotic organism are placed in one domain not in kingdom

@EXAMLIX

36) Find the correct statements about lichen :-
(A) Lichens are mutually useful association between algae and fungi.
(B) Algae prepare food for fungi and the fungi provide shelter to its partner.
(C) Lichens are very good pollution indicators.
(D) The alga is phycobiont and the fungus is mycobiont

(1) A & B
(2) B & C
(3) A, B & D
(4) A, B, C & D

37) Archaebacteria differ from other bacteria in having a :-

(1) Different cell wall and cell membrane structure
(2) Different cellular organization
(3) Parasitic nature
(4) Both (2) and (3)

38) Choose the wrong statement :

(1) Yeast is unicellular and useful in fermentation

(2) Penicillium is multicellular and produces antibiotics

(3) Neurospora is used in the study of biochemical genetics

(4) Morels and truffles are poisonous mushrooms

39) Read the following given words :-
Basidiopore, mushroom, Smut disease, *Claviceps*, late blight of potato.
How many of them related to Basidiomycetes ?

(1) 2

(2) 3

(3) 4

(4) 5

40) Which of the following is **correct** about viruses?

(1) They have free DNA without protein coat.

(2) They have abnormally folded proteins.

(3) They have free RNA without protein coat.

(4) They have DNA or RNA with protein coat.

41) Heterotrophic, Eukaryotic, Multicellular organisms without cell wall are included in which kingdom ?

(1) Fungi

(2) Plantae

(3) *Animalia*

(4) Both (1) & (3)

@EXAMLIX

42) In which year Whittaker proposed the five kingdom classification?

(1) 1960

(2) 1959

(3) 1969

(4) 1979

43) Bacteria participate in -

(1) Making curd from milk

(2) N_2 - fixation in legume roots

(3) Disease causing in plants, animals and human beings

(4) All of these

44) How many of the given below are related with Archaebacteria ? Gut of ruminants, Biogas production, Halophiles, Thermoacidophilous , Heterocyst , Blooms in polluted water bodies, Chlorophyll a

(1) 2

(2) 3

(3) 4

(4) 5

45) Which option is related with only monera and not related with other kingdom of R.H. Whittaker?

(1) Decomposer nature

(2) Presence of cell wall and cell membrene

(3) Well develop tissue system

(4) Nitrogen fixation ability

ZOOLOGY

1) What will happen if the stretch receptors of the urinary bladder wall are totally removed ?

(1) There will be no micturition

(2) Urine will not collect in the bladder

(3) Micturition will continue

(4) Urine will continue to collect normally in the bladder

@SAMLIX

2) Consider the following statements:
A. Podocytes are associated with the inner wall of Bowman's capsule.
B. Filtration slits occur between the processes of podocytes.
C. The glomerulus is a tuft of capillaries formed from the efferent arteriole.
D. Blood leaves the glomerulus through the efferent arteriole.
Which of the above statements are correct?

(1) A and C only

(2) B, C and D only

(3) A, B and D only

(4) A, B, C and D

3) Ammonia produced by metabolism is converted into the ....A.... in the ....B... in ureotelic and released into the blood, which is filtered and excreted out by ...C...

(1) A-uric acid, B-spleen, C-kidney

(2) A-uric acid, B-liver, C-kidney

(3) A-urea, B-liver, C-kidney

(4) A-urea, B-spleen, C-kidney

4) Match the following parts of a nephron with their function:

(a) Descending limb of Henle's loop	(i) Reabsorption of salts only

(b) Proximal Convoluted tubule	(ii) Reabsorption of water only
(c) Ascending limb of Henle's loop	(iii) Conditional reabsorption of sodium ion and water
(d) Distal convoluted tubule	(iv) Reabsorption of ion, water and organic nutrients.

Select the correct option from the following :

(1) (a)-(i), (b)-(iii), (c)-(ii), d-(iv)

(2) (a)-(ii), (b)-(iv), (c)-(i), d-(iii)

(3) (a)-(i), (b)-(iv), (c)-(ii), d-(iii)

(4) (a)-(iv), (b)-(i), (c)-(iii), d-(ii)

5) **Assertion (A) :** Ammonia is the excretory product of aquatic amphibian.
Reason (R) : Ammonia is readily soluble in water and requires large amount of water for excretion.

(1) Both (A) and (R) are correct and (R) is the correct explanation for (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation for (A)

(3) (A) is correct but (R) is incorrect

(4) (A) is incorrect but (R) is correct

6) Which part of nephron is impermeable to water?

(1) Descending limb of henle's loop

(2) Ascending limb of henle's loop

(3) PCT

(4) DCT

7) Kidney stones are produced due to :-

(1) Precipitation of cholesterol.

(2) Deposition of sand particles.

(3) Blockage of fat.

(4) Crystallisation of oxalates.

8) To keep the osmolarity, small amount of urea moves into medullary interstitium from -

(1) PCT

(2) Thin segment of loop of Henle

(3) Thick segment of loop of Henle

(4) Distal part of collecting duct

9)

@EXAMLY

Which of the following does not play a role in increasing osmolarity towards inner medullary interstitium ?

(1) Na^+

(2) Urea

(3) Cl^-

(4) K^+

10)

Correct match list-1 with list-2

List-1		List-2	
(A)	Glycosuria	(i)	Inflammation of glomeruli of kidney
(B)	Ketonuria	(ii)	Presence of glucose in urine
(C)	Glomerulonephritis	(iii)	Presence of blood in urine
(D)	Haematuria	(iv)	Presence of ketone bodies of urine

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(ii), B-(iv), C-(iii), D-(i)

(3) A-(ii), B-(iv), C-(i), D-(iii)

(4) A-(i), B-(ii), C-(iv), D-(iii)

11)

Which part of the tubule have highest osmolarity

(1) Collecting duct

(2) PCT

(3) DCT

(4) At the base of loop of henle

12) In the counter current mechanism the NaCl is transported from
(a) Ascending limb of Henle's loop
(b) Descending limb of Henle's loop
(c) Ascending portion of vasarecta
(d) Descending portion of vasarecta
(e) Medullary interstitium of kidney

(1) $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e$

(2) $a \rightarrow d \rightarrow c \rightarrow e$

(3) $a \rightarrow d \rightarrow c \rightarrow b \rightarrow e$

(4) $a \rightarrow c \rightarrow d \rightarrow e$

13) Which one of the following is the ultimate method in the correction of acute renal failure ?

@EXAMLIX

(1) Transplantation of a functioning kidney from a donor to the host.

(2) Transplantation of a non-functioning kidney from a donor to the host.

(3) Drug assisted purification of blood.

(4) Hemodialysis.

14) If the permeability of the collecting duct to water decreases significantly, the most like consequence is :

(1) Formation of highly concentrated urine

(2) Formation of dilute urine

(3) Increase urea synthesis

(4) Increase glomerular filtration

15) A person passes much urine and drinks much water, but his blood glucose level is normal. This condition may be the result due to :

(1) a reduction in insulin secretion from pancreas

(2) a reduction in vasopression secretion from Hypothalamus

(3) a fall in the glucose concentration in urine

(4) an increase in secretion of glucagon

16) **Assertion :** Mammals have ability to produce a concentrated urine.
Reason : Countercurrent mechanism helps to maintain a concentration gradient in the medullary interstitium.

(1) Assertion is correct, reason is correct & reason is correct explanation of assertion.

(2) Assertion is correct, reason is correct but reason is not correct explanation of assertion.

(3) Assertion is correct, reason is not correct.

(4) Both assertion & reason are incorrect.

17)

The outline of principal event of urination is given below in unorder manner –
I. Stretch receptors on the wall of urinary bladder send signal to the CNS
II. The bladder fills with urine and becomes distanded
III. Micturition
IV. CNS passes on motor signals to initiate the contraction smooth muscles of bladder and simultaneous relaxation of urethral Sphincter.

The correct order of steps for urination is –

(1) I → II → III → IV

(2) IV → III → II → I

(3) II → I → IV → III

(4) III → II → I → IV

18)

@EXAMLIX

Internal section of Kidney shows extension of cortex into medullary regions between the pyramids and is called

(1) Ducts of Bellini

(2) Major calyx

(3) Minor calyx

(4) Columns of Bertini

19) JG cells are activated in response to ?
(A) Fall in GFR
(B) Decrease in Glomerular blood pressure
(C) Increase in Blood pressure of Renal artery

(1) A & C

(2) A & B

(3) B & C

(4) All of the above

20) Which one of the following is not a part of a renal pyramid?

(1) Loops of Henle

(2) Peritubular capillaries

(3) Convoluted tubules

(4) Collecting ducts

@EXAMLIX

21) During hemodialysis process
I. blood drained from a convenient artery and anticoagulant is added (heparin).
II. removal of nitrogenous waste from blood.
III. blood is passed through a coiled porous cellophane membrane of tube bathing in dialysis fluid.
IV. blood is mixed with antiheparin and passed into vein.
Arrange the steps :-

(1) I → II → III → IV

(2) IV → III → II → I

(3) I → III → II → IV

(4) I → IV → II → III

22) ANF is secreted by the

(1) Atrial wall when the venosus return is high

(2) Atrial wall when the venosus return is low

(3) JG cell when the venosus return is low

(4) Nephron wall when the venosus return is high

23)

Match the following animals with the excretory organ related to them.

Animal		Excretory organ	
(i)	Earthworm	(a)	Nephridia
(ii)	Cockroach	(b)	Protonephridia
(iii)	Prawn	(c)	Malpighian Tubules
(iv)	*Amphioxus*	(d)	Green glands

(1) (i)-a, (ii)-c, (iii)-d, (iv)-b

(2) (i)-a, (ii)-b, (iii)-c, (iv)-d

(3) (i)-d, (ii)-c, (iii)-b, (iv)-a

(4) (i)-c, (ii)-d, (iii)-a, (iv)-b

24) **Assertion -** Nitrogenous waste is absent in dialysing fluid.
Reason - Nitrogenous waste diffuse from blood to dialysing fluid on the basis of concentration gradient.

(1) Both assertion and reason are true and the reason is the correct explanation of assertion.

(2) Both assertion and reason are true but the reason is not the correct explanation of assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion and reason are false.

25) Which is reabsorbed actively in nephron?
(a) Glucose (b) Cl^-
(c) K^+ (d) Nitrogenous wastes

(1) a and b only

(2) Only a

(3) a and d only

(4) a, b and d

26) Number of nephrons per kidney is about:

(1) 10,000

(2) 1 lakh

(3) 10 million

(4) 1 million

27) Match the following?

	Column-I		Column-II
A	PCT	i	Selective secretion of H^+ and K^+ ions
B	DCT	ii	70-80 percent of electrolytes and water reabsorbed

@EXAMLIX

C	Loop of Henle	iii	Allow passage of small amount of urea
D	Ascending limb	iv	Concentration of urine
E	Collecting duct	v	Permeable to salts

(1) A-i ; B-ii ; C-iii ; D-iv ; E-v
(2) A-i ; B-iii; C-iv ; D-ii ; E-v
(3) A-ii ; B-i ; C-iv ; D-v ; E-iii
(4) A-ii ; B-iii ; C-iv ; D-v ; E-i

(1) 1

(2) 2

(3) 3

(4) 4

28) In collecting duct selective secretion occurs of some ions. These are :-

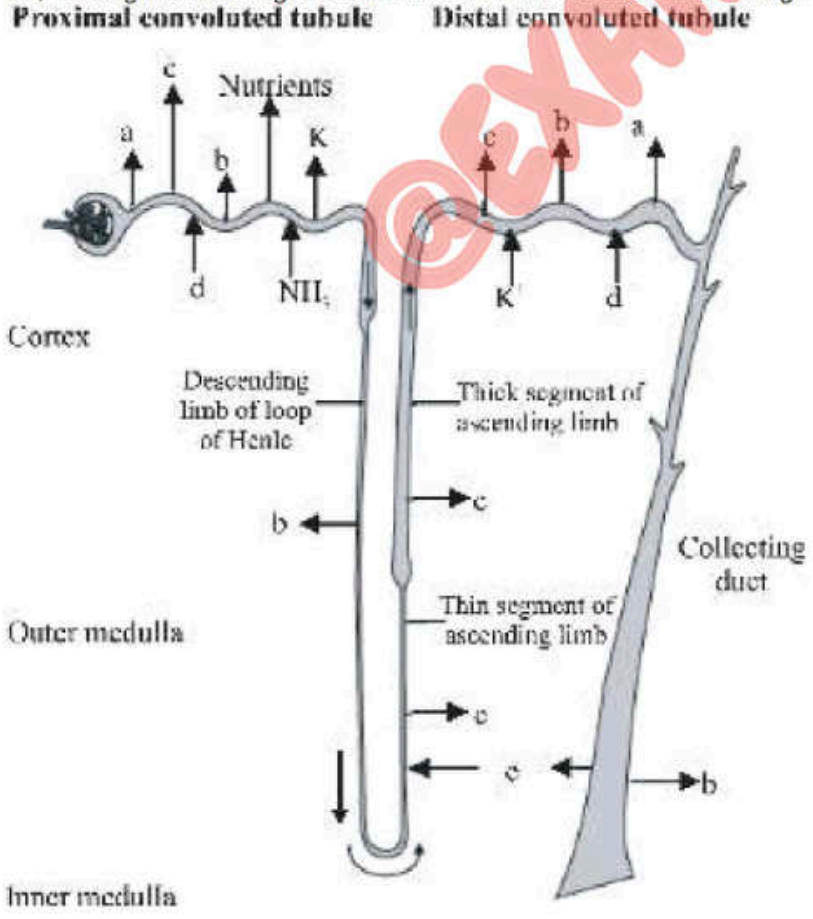
(1) NH_3 only

(2) H^+ & K^+

(3) NH_3 & K^+ only

(4) NH_3 & H^+ only

29) Recognise the figure and find out the correct matching.



(1) b-H^+, d-H_2O, c-Urea, a-NaCl, e-HCO_3^-

(2) c-H^+, b-H_2O, d-Urea, e-NaCl, a-HCO_3^-

@EXAMLIX

(3) d-H^+, b-H_2O, e-Urea, c-NaCl, a-HCO_3^-

(4) a-H^+, c-H_2O, b-Urea, d-NaCl, e-HCO_3^-

30) Select the **correct** statements.
(a) Angiotensin II activates the cortex of adrenal gland to release aldosterone.
(b) Aldosterone leads to increase in blood pressure.
(c) ANF acts as a check on renin-angiotensin mechanism.
(d) ADH causes vasodilation.
(e) Vasopressin is released from adenohypophysis.
Choose **the most appropriate answer** from the options given below :

(1) (a), (b) and (e) only

(2) (c), (d) and (e) only

(3) (b), (c) and (d) only

(4) (a), (b) and (c) only

31) Large amount of water is ....A... from collecting duct to produce ...B... urine. This segment allows passage of small amounts of ...C... into interstitium of medulla to keep up the osmolarity. Here, A, B and C refers to

(1) A- secreted, B-dilute, C-sugar

(2) A- secreted, B-dilute, C-NH_3

(3) A- secreted, B-dilute, C-urea

(4) A- reabsorbed, B-concentrated, C-urea

32) Choose the correct pathway of urine flow in human beings :

(1) Henle's loop → PCT → Collecting duct → Ureter → Urinary bladder → Urethra

(2) DCT → PCT → Bowman's capsule → Urinary bladder → Urethra

(3) PCT → DCT → Bowman's capsule → Urinary bladder → Urethra

(4) Henle's loop → DCT → Collecting duct → Ureter → Urinary bladder → Urethra

33) Removal rate of carbon dioxide by lungs is approximately.

(1) 125 ml/minute

(2) 200 ml/minute

(3) 280 ml/minute

(4) 325 ml/minute

34) Minimum reabsorption of H_2O occurs in which segment of nephron?

(1) Descending limb of loop of Henle

(2) Ascending limb of loop of Henle

(3) Distal convoluted tubule

(4) Collecting duct

35) How much amount of blood is filtered by the kidneys per minute :-

@EXAMLIX

(1) 180 litres

(2) 125 ml

(3) 1100-1200 ml

(4) 650 ml

36) If Henle's loop is absent from mammalian nephron which of the following event is likely to be expected?

(1) There will be no urine formation.

(2) There will be hardly any change in the quality and quantity of urine formed.

(3) The urine will be more concentrated.

(4) The urine will be more dilute.

37) Juxta-glomerular apparatus is formed by cellular modification in the :–

(1) afferent arteriole and DCT

(2) efferent arteriole and PCT

(3) afferent arteriole and PCT

(4) efferent arteriole and DCT

38) (A) Concentration of urine is function of Juxta-medullary Nephron.
(B) Afferent arteriole has more diameter than efferent arteriole.
(C) Rennin is a vasoconstrictor
Read above statements carefully & choose correct option

(1) A is correct while, B and C are incorrect

(2) B is correct while, A and C are incorrect

(3) A, B are correct while, C is incorrect

(4) A, C are correct while, B is incorrect

39) Select the statement that most accurately integrates the NCERT description of urine and its clinical use:

(1) Urine is a slightly alkaline fluid used primarily to check for liver cirrhosis.

(2) The presence of any amount of urea in urine is considered a sign of kidney failure.

(3) The 25-30g of urea excreted daily is a byproduct of protein metabolism, and its analysis helps in diagnosing metabolic disorders.

(4) Micturition is an entirely endocrine-controlled process with no neural involvement.

40) In response to decrease in blood volume and blood pressure, which of the following do not occurs ?

(1) Secretion of Renin

(2) Secretion of Aldosterone

(3) Secretion of Vassopressin

(4) Secretion of A.N.F

@EXAMLIX

41)

Read the following statements and choose which is not incorrect ?
(A) Atrial natriuretic factor can cause lower in blood pressure
(B) Angiotensin II is a vasoconstrictor
(C) ANF is a vasodilator
(D) ADH is a vasodilator

(E) ADH can cause diuresis

(1) A, B only

(2) A, B, C only

(3) C, D, E only

(4) C, E only

42) An increase in the body fluid volume can switch off the ...A... and ...B... the ADH release.
In this way complete the ...C... .
Choose the correct option for A, B and C.

(1) A-osmoreceptors, B-increase, C-feedback

(2) A-osmoreceptors, B-suppress, C-feedback

(3) A-kidney filtration, B-increase, C-feedback

(4) A-kidney filtration, B-suppress, C-feedback

43) Which of the following statements are not incorrect?
(i) Liver, the largest gland in our body secretes cholesterol, degrade steroid hormones, vitamins and drugs
(ii) Sebaceous glands eliminate certain substances like urea, sterols and sebum
(iii) Sweat produced by sweat glands is a oily fluid containing NaCl
(iv) In uremic patients urea and glucose can be removed by a process called hemodialysis

(1) i, ii, iii

(2) ii, iii, iv

(3) i only

(4) i, ii, iv

44) Which of the following statements are correct :-
(a) vasa recta runs parallel to the henle's loop in juxtamedullary nephrons
(b) cortical nephron has no or highly reduced vasa recta
(c) vasa recta, peritubular capillaries, glomerulus all have blood
(d) blood vessel leading to glomerulus is called efferent arteriole

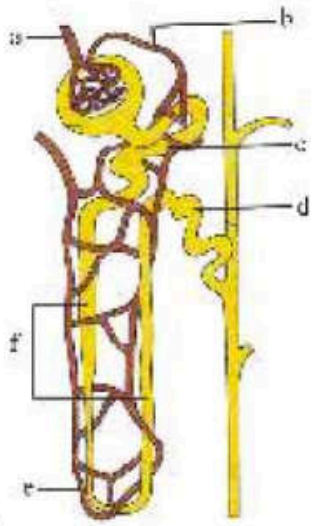
(1) only 'a'

(2) a and b only

(3) a, b and c

(4) all are correct

@EXAMLIX



45) Recognise the figure and find out the correct matching :-

(1) c-PCT, d-DCT, a-afferent arteriole, b-efferent arteriole, f-henle's loop, e-vasa recta

(2) d-PCT, c-DCT, b-afferent arteriole, a-efferent arteriole, e-henle's loop, f-vasa recta

(3) c-PCT, d-DCT, b-afferent arteriole, a-efferent arteriole, f-henle's loop, e-vasa recta

(4) d-PCT, c-DCT, a-afferent arteriole, b-efferent arteriole, e-henle's loop, f-vasa recta

@EXAMLIX

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	3	1	4	2	1	2	1	1	3	3	2	3	1	3	2	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	2	1	3	1	3	1	2	1	4	4	2	4	4	3	3	1	3	1
Q.	41	42	43	44	45															
A.	3	1	4	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	2	2	2	3	2	2	1	3	1	1	4	3	2	3	4	4	1	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	4	1	3	2	1	3	4	4	3	2	1	3	1	2	1	4	4	4	1
Q.	86	87	88	89	90															
A.	4	4	3	2	2															

BOTANY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	2	4	2	1	3	3	1	1	4	2	3	1	4	3	2	2	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	3	2	3	3	4	2	3	1	3	4	3	1	4	1	4	2	4
Q.	131	132	133	134	135															
A.	3	3	4	3	4															

@EXAMLIX

ZOOLOGY

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155
A.	1	3	3	2	1	2	4	4	4	3	4	2	1	2	2	1	3	4	2	3
Q.	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175
A.	3	1	1	1	2	4	3	2	3	4	4	4	2	2	3	4	1	3	3	4
Q.	176	177	178	179	180															
A.	2	2	3	3	1															

SOLUTIONS

PHYSICS

1) **Question Explanation:**
Find the distance for stable equilibrium.
Concept:
Equilibrium occurs where Net Force is zero.
Solution:
At equilibrium, the change in potential energy (U) with respect to distance (r) is zero. We differentiate the energy formula and solve for r.
Given $U = \frac{A}{r^3} - \frac{B}{r^2}$ (for the intended answer):

$$F = -\frac{dU}{dr} = -\left[\frac{-3A}{r^4} - \frac{-2B}{r^3}\right] = 0$$

$$\frac{3A}{r^4} = \frac{2B}{r^3}$$

$$r = \frac{3A}{2B}$$

Final Answer:
Option (2)

@EXAMLIX

2) **Explain Question:**
Question ask to find acceleration of the rocket.

Concept:
Variable mass space

Solution:
$\frac{dM}{dt}$ =0.1 kg/s, v_{gas} = 50 m/s,
mass of the rocket = 2 kg, Mv = constant

$$-v\frac{dM}{dt} + M\frac{dv}{dt} = 0$$

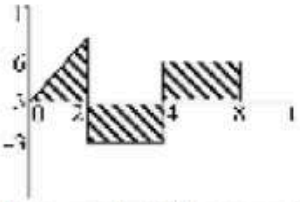
$$\frac{dv}{dt} = \frac{1}{M}v\frac{dM}{dt}$$

or acceleration $= \frac{1}{2} \times 50 \times 0.1 = 2.5\ m/s^2$.

Final Answer: option(3)

3) $\frac{mg}{m(g-a)} = \frac{3}{2}$ $a = \frac{g}{3}$

4) **Explain the Question :** Find change in momentum form 0 to 8sec.
Concept : This question is based on change in momentum (F vs t curve).
Solution :



Area under F-t curve gives change in momentum (ΔP)

$\Delta P = \int F dt$

$\Delta P = \frac{1}{2} \times 6 \times 2 + (-3)(4-2) + 3 \times (8-4)$

$\Delta P = 6 - 6 + 12$

$\Delta P = 12$ Ns

Final Answer : (3)

5) $F = \frac{vdm}{dt}$

$ma = \frac{vdm}{dt}$

$2a = 5 \times 1 \Rightarrow a = 2.5\ m/s^2$

6) Δp = Area of force time-graph = 0

7) **Explanation Question:** We need to find the maximum upward acceleration of a monkey on a rope on the basis of given data.

Concept: Newton's Second Law of Motion.

Solution:

$T_{max} - mg = ma_{max}$

$600 - (40 \times 10) = 40 \times a_{max}$

$200 = 40a_{max}$

$a_{max} = \frac{200}{40} = 5m/s^2$

Final answer: Option (2)

8) Useful P = mgh/t = 100×10×10/5 = 2000 W. Input P = 2000/0.60 ≈ 3333 W = 3.33 kW.

9)

Impulse = $\Delta p = m(v_2 - v_1)$

$= 70\left(\sqrt{2gh} - 0\right)$

$= 70\left(\sqrt{2 \times 9.8 \times 0.8} - 0\right)$

$= 70 \times 3.96$

$= 277.2$ kgm/sec

10)

@EXAMLIX

Question Explanation : Given the **momentum (P) vs. time (t) graph**, which shows a decrease in momentum from P = 20 N-s at t = 4 s to P = 0 N-s at t = 8s, find the **magnitude of the negative force** acting on the particle during this interval.

Concept : This question is based on Force = rate of change of linear momentum

Formula : $F = \frac{dP}{dt}$ = Slope of the P – t graph

Solution : A negative force corresponds to a **negative slope** on the P – t graph, which occurs during the time interval from t = 4s to t = 8s.

$$F = \frac{P_2 - P_1}{t_2 - t_1} = \frac{0\ N \cdot s - 20N \cdot s}{8s - 4s} = \frac{-20}{4} N$$

$F = -5\ N$

Magnitude of Force = $|-5N| = 5N$

Final answer : option (1)

$$F\cos\theta = \mu N$$

$$F\sin\theta + N = mg$$

$$\Rightarrow F = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$$

$$F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

$$= \frac{\frac{1}{\sqrt{3}} \times 10}{\frac{2}{\sqrt{3}}} = 5$$

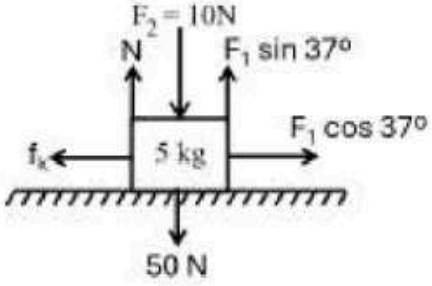
11)

@EXAMLIX

12) **Question Explanation:** Calculate acceleration of block on rough Horizontal surface.

Concept: Motion of a block on rough horizontal surface.

Solution: Resolving the force F_1 into its components and drawing the FBD for the block



In vertical direction :

$N + F_1 \sin 37^\circ = F_2 + 50$

$N + 60 \times \frac{3}{5} = 10 + 50 \Rightarrow N = 24$ Newton

In horizontal direction :

$F_{net} = F_1 \cos 37^\circ - f_k$ (f_k = kinetic friction = $\mu_k N$)

$ma = 60 \times \frac{4}{5} - 0.5 \times 24$

$5a = 48 - 12$

$\Rightarrow a = \frac{48-12}{5} = \frac{36}{5} = 7.2\ m/s^2$

Final Answer: Option (3)

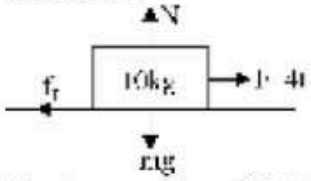
13)

F=ma and slope of v-t graph gives accelaration.

14) **Explain the Question :** Find minimum value of t to move the block.

Concept : This question is based on motion on rough horizontal plane.

Solution :



Maximum value of friction = μmg = 0.04 × 10 × 10]

$f_r = 40$ N

block start moving when applied force equal or greater than maximum value of friction.

$f_r = F$

$F = 4t = 40$

$t = \frac{40}{4} = 10\ sec$

Final Answer : (2)

15)

Explain the Question : We need to find the horizontal velocity of the block shown in the figure, from given data.

Concept : This question is based on constraint motion.

Solution :

$v' \sin\theta = v$

$v' = \frac{v}{\sin\theta}$

Final answer : 3

16) $P = Fv = m\,v\,dv/dt = P_0 \Rightarrow v\,dv = (P_0/m)\,dt$. Integrate: $v^2 \propto t \Rightarrow v \propto \sqrt{t}$.

17) **Explaining :** A particle moves with a velocity $\vec{v} = (5\hat{i} - 3\hat{j} + 6\hat{k})\ ms^{-1}$ under the influence of a constant force $\vec{F} = (10\hat{i} + 10\hat{j} + 20\hat{k})\ N$. The instantaneous power applied to the particle is:-

Concept : This question is based on power

Solution : $P_{inst} =$

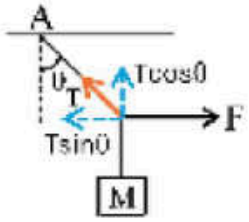
$\vec{F}.\vec{V} = (10\hat{i} + 10\hat{j} + 20\hat{k}) \cdot (5\hat{i} - 3\hat{j} + 6\hat{k})$

$= 50 - 30 + 120 = 170 - 30 = 140$ w

Final Answer : (3)

@EXAMLIX

18) **Question Explanation :** Find the tension in string AB when a mass is suspended and a horizontal force F is applied, causing the string to make an angle θ with the vertical. Find the tension in terms of F and θ, given the system is in equilibrium.
Concept: Equilibrium
Solution:



When a system is in equilibrium, the sum of forces in each direction (horizontal and vertical) must be zero.

$T \sin\theta = F \Rightarrow T = \frac{F}{\sin\theta}$
Final Answer: Option (2)

19) **Explain** :- The problem shows a system in equilibrium with two weights, W_1 and W_2 suspended by strings at angles. We're given W_1 = 300 N and need to find W_2.

Concept :- Force balance for equilibrium.

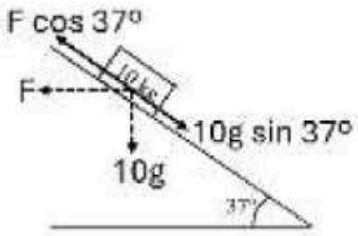
Formula : $\Sigma\vec{F}x = 0$(sum of horizontal forces is zero)
$\Sigma\vec{F}y = 0$ (sum of vertical forces is zero)

Tension components :-
$Tx = T\cos(\theta)$
$Ty = T\sin(\theta)$

Calculation : Horizontal equilibrium $(\Sigma\vec{F}x = 0)$
$T_1\cos(37°) = T_2\cos(37°)$
$T_1 = T_2$
Vertical Equilibrium $(\Sigma\vec{F}y = 0)$
$T_1\sin(37°) = W_1 = 300N$
$W_2 = T_2\sin(37°) = T1\sin(37°) = 300$ N

Final Answer : (4)

20) **Question Explanation:** We have to calculate external force F.
Concept: Equilibrium.
Solution:



@EXAMLIX

In equilibrium:

$F\cos 37° = mg \sin 37°$

$F \times \frac{4}{5} = 10 \times 10 \times \frac{3}{5}$

$F = 75$ N

Final answer: Option (3)

21)

Question Explanation :

We need to find the readings of the spring balances A and B shown

Concept :

Equilibrium under tension force involving springs.

Solution :

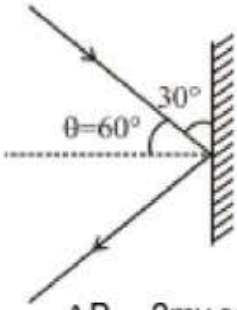
Reading of spring balance = Tension in each spring
= 4 kg

Final answer : Option (3)

22) **Explain Question:**
Question ask to find fore experienced by the wall is

Concept:
impulse & average force

Solution:



$$F = \frac{\Delta P}{\Delta t} = \frac{2mv\cos\theta}{t}$$

$$= \frac{2 \times (100 \times 10^{-3}) \times 10 \times \cos 60°}{0.1}$$

$\Rightarrow 10$ N

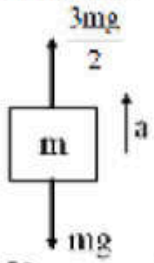
Final Answer: option(1)

23) **Explain the question :** To find the acceleration of the block.
Concept: This question is based on motion of bodies connected with pulley systems without spring

@EXAMLIX

Formula: Net force = mass × acceleration

Solution:



$$\left[\frac{3mg}{2} - mg\right] = ma$$

$$\Rightarrow a = \left(\frac{g}{2}\right) \text{ upward}$$

Final answer : 2

24) **Question Explanation:**

Calculating contact forces between three blocks.

Concept:

System acceleration and Free Body Diagrams.

Solution:

Total mass = 3 + 2 + 1 = 6kg

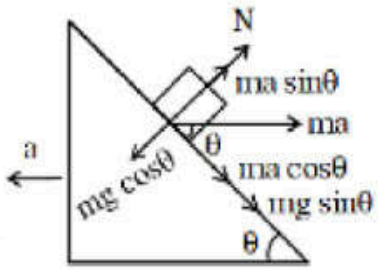
Acceleration (a) = $\frac{12N}{6kg} = 2m/s^2$

f_2 (force on 3kg block) = $m^3 \times a = 3 \times 2 = 6N$

f_1 (force between 2kg and 1kg) pushes (3kg + 2kg)

$f_1 = (3 + 2) \times a = 5 \times 2 = 10N$

Final Answer: Option (1)



25)

$N + ma \sin\theta = mg \cos\theta$

If the block falls freely, N = 0

$ma \sin\theta = mg \cos\theta$

$a = g \cot\theta$

26)

Explain the Question : Column I lists types of work done by forces: conservative, non-conservative, and total. Column II lists corresponding energy changes: ΔK (kinetic), –ΔU (negative potential), and ΔE (total mechanical). We must match each correctly.

@EXAMLIX

Concept : This question is based on Work-energy relations for different force types.

Solution :
A. Conservative force: $W = -\Delta U$
B. Non-conservative force: $W = \Delta K - (-\Delta U) = \Delta K + \Delta U$
C. All forces: $W = \Delta K$

A → $-\Delta U \Rightarrow Q$
B → Change in total mechanical energy = $\Delta E \Rightarrow R$
C → Work by all forces = $\Delta K \Rightarrow P$

Final answer : Option (1)

27) Let max elongation of spring be x
Change in PE of block = PE stored in spring

$$mgx = \frac{1}{2}kx^2$$

$$\boxed{x = \frac{2mg}{K}}$$

28) **Explain Question:**

Determine the coefficient of friction μ if a particle loses $\frac{3}{4}$ of its kinetic energy in time t^0.

Concept:
Kinematic equation v = u + at and Kinetic Energy relation.

Solution:

Initial Kinetic Energy $K_0 = \frac{1}{2}mv_0^2$

Final Kinetic Energy $K = K_0 - \frac{3}{4}K_0 = \frac{1}{4}K_0$

$$\frac{1}{2}mv^2 = \frac{1}{4}\left(\frac{1}{2}mv_0^2\right)$$

$$v = \frac{v_0}{2}$$

Using v = u + at where retardation a = -μg:

$$\frac{v_0}{2} = v_0 - \mu g t_0$$

$$\mu g t_0 = \frac{v_0}{2}$$

$$\mu = \frac{v_0}{2gt_0}$$

Final Answer: Option (1)

29)

Explain Question:
A 1kg block sides at 4 m/s on a smooth track, then hits a spring [k = 2N/m] on a rough surface [m = .2].

@EXAMLIX

Concept:
Conservative 2 non-conservative forces.

Solution:
From energy conservation,
K . E of block = U_s + loss by frication.
$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 + f_r x$
$\frac{1}{2}(1)(4)^2 = \frac{1}{2}[2][x]^2 + [0.2(1)(g)][x]$
$8 = x^2 + 2x$
$(x + 4)(x - 2) = 0$
$x = -4, +2$
$x = 2m$

Final answer: Option (2)

30) $\vec{F} . \vec{d} = W$
$(3\hat{i} + c\hat{j} + 2\hat{k}).(-4\hat{i} + 2\hat{j} - 3\hat{k}) = 6$
$-12 + 2C - 6 = 6$
$C = 12$

31) $\frac{1}{2}kx_{max^2} = Mgh$
$\frac{1}{2} \times 400 \times x_{max^2} = \frac{40}{100} \times 10 \times 5$
$x_{max} = \frac{1}{10}m = 10cm$

32)

$W = \Delta KE = Area$
$\frac{1}{2}MV^2 = \frac{1}{2}(4 + 12) \times 10$
$V = 40\ m/s^2$

33) $Mg(h + x) - F_{AV}X = 0 - 0$
$F_{AV} = Mg\left(\frac{h}{x} + 1\right)$

34) $U = \frac{1}{2}K(2 \times 10^{-2})^2$
$U' = \frac{1}{2}K(10 \times 10^{-2})^2 = \frac{1}{2}K(2 \times 5 \times 10^{-2})^2 = U(5)^2 = 25U$

35) $W_{ext} = U_f - U_i$

@EXAMLIX

$$= O - \left[-\frac{M}{3} \times g \times \frac{L}{2 \times 3}\right]$$

$$W_{ext} = \frac{MgL}{18}$$

36) **Explanation:** In this question we have to find the Change in Kinetic energy if Change in momentum is given.

Concept: The Relation between Kinetic energy and momentum is given by

$$k = \frac{p^2}{2M}$$

where k → kinetic energy

P → linear momentum

M → mass of the body

$$\% \text{ change in KE} = \left(\frac{k_{final} - k_{initial}}{k_{initial}}\right) \times 100$$

Formula: Explained in Concept Part

Calculation: Let the initial KE be K_i and initial momentum be P_i

$\therefore k_i = \frac{p_i^2}{2M}$ -----(1) (M = Constant)

Now since momentum is increased by 100%

$\therefore$ New momentum $P_f = P_i + 100\%$ of $P_i = 2P_i$

$\therefore k_f = \frac{(P_f)^2}{2M} \Rightarrow k_f = \frac{(2P_i)^2}{2M} \Rightarrow \frac{4P_i^2}{2M}$ (M = Constant)

$\therefore \%$ change in KE $= \frac{(k_f - k_i)}{k_i} \times 100$

$$\Rightarrow \left(\frac{\frac{4P_i^2}{2M} - \frac{P_i^2}{2M}}{\frac{P_i^2}{2M}}\right) \times 100 \quad \Rightarrow \frac{\frac{P_i^2}{2M}(4-1)}{\frac{P_i^2}{2M}} \times 100$$

= 300 %

Conclusion: The correct answer is option 3.

37) **Explain Question:**

Determine the ratio of kinetic energy to potential energy for a stone at a specific height.

Concept:

Law of Conservation of a Mechanical Energy

Solution:

Total Energy at max height H is $E = mgh$

At height $h' = \frac{4h}{5}$, potential Energy $U = mgh' = mg\left(\frac{4h}{5}\right)$

$U = \frac{4}{5}mgh$

kinetic Energy $K = E - U = mgh - \frac{4}{5}mgh \; K = \frac{1}{5}mgh$

Ratio $\frac{U}{K} = \frac{(1/5)\,mgh}{(4/5)\,mgh}$

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$$\frac{K}{U} = \frac{1}{4}$$

Final answer: Option (3)

38)

Explain Question:
Calculate the work done by the tension in a string pulling a block of mass m upward with a constant acceleration g/2.

Concept:
Works = Force × Displacement × cos(θ) , where Tension T = m(g + a).

Solution:
Net force equation for upward motion: T – mg = ma
Given acceleration a = g / 2
T = mg + m(g / 2) = 3mg / 2
Work done by tension W_t = T × h × cos(0°)
W_t = (3mg / 2) × h
W_t = +3mgh / 2

Final answer: Option (1)

39)

$$W.D = 3\hat{i}.4\hat{i} + 4\hat{j}.3\hat{j}$$
$$= 12 + 12$$
$$= 24 \text{ Joule}$$

@EXAMLIX

40)

Explain Question:
Calculate the work done by a constant horizontal force F in pulling a block along a circular arc from point A to point B.

Concept:
Work done by a constant force is the product of the force and the displacement in the direction of the force.

Solution:
Horizontal displacement x is the distance from the vertical axis at point A to point B.
At point A, the horizontal position is 0.
At point B, the horizontal position relative to the center is R × sin(30°)
Net horizontal displacement Δx = R × sin(30°) – 0
Δx = R × (1/2) = R / 2
Work done W = Force Displacement in direction of force
W = F × Δx, W = F × (R / 2) = $\frac{FR}{2}$

Final answer: Option (2)

41)

Explain Question:
Identify the points on a potential energy-distance graph where the force acting on the particle is zero.

Concept:
Force is related to potential energy by $F = -dU / dx$ (the negative slope of the U – x graph).

Solution:
Force F is zero where the slope of the potential energy graph dU/dx is zero.
On the given graph, the slope is zero at local maxima and minima.
Point B is a local maximum (slope= 0)
Point C is a local minimum (slope= 0)
Therefore, the force is zero at both points B and C.

Final answer: Option (3)

@EXAMLIX

42) **Explain Question:**
Determine the type of equilibrium at the point x = 2 for a particle with the potential energy function $U(x) = x^2 - 4x$.

Concept:
Equilibrium type is determined by the first and second derivatives of potential energy.

Solution:
$U(x) = x^2 - 4x$ $dU/dx = 2x - 4$
At x = 2: $dU/dx = 2(2) - 4 = 0$ (Equilibrium point)
$d^2U / dx^2 = 2$
Since $d^2U / dx^2 > 0$ (positive), the potential energy has a local minimum at x = 2
A local minimum in potential energy corresponds to stable equilibrium.

Final answer: Option (1)

43) **Explanation :** The block starts at rest at height h = 2m and eventually comes to rest on the horizontal track. Mechanical energy loss equals the negative work done by friction.
Concept : This question is based on $W_{total} = \Delta K = 0$
$W_{gravity} + W_{friction} = 0$
Solution :
$m = 0.2$ kg, $h = 2$ m, $g = 10$ m/s²
$W_{gravity} = m \cdot g \cdot h = 0.2 \cdot 10 \cdot 2 = 4$ J
$4 + W_{friction} = 0$
$W_{friction} = -4$ J
Final Answer : (4)

44) **Explain the Question :** Compare work done on springs stretched by equal force.

Concept : This question is based on Work done in stretching a spring.

Solution: For a spring, $x = \frac{F}{k}$.

Work done $W = \frac{1}{2}kx^2 = \frac{1}{2}k\left(\frac{F}{k}\right)^2 = \frac{F^2}{2k}$

Spring 1 : k_1, $W_1 = \frac{F^2}{2k_1}$

Spring 2 : k_2, $W_2 = \frac{F^2}{2k_2}$

Since $k_1 > k_2$, $W_1 < W_2$.

Hence, more work is done on the softer spring (spring 2).

Final answer : 3

45)

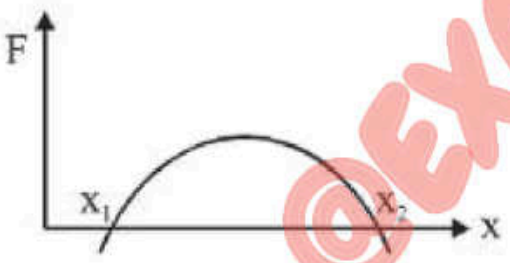
QUESTION EXPLANATION The force acting on a body moving along the x-axis varies with position as shown. Force is positive left of x_1, negative between x_1 and x_2, and positive right of x_2. We must find the point of stable equilibrium.

CONCEPT Equilibrium and stability

FORMULA

Stable equilibrium: Force changes sign from negative to positive with restoring tendency

SOLUTION



At x_1: Force changes from positive (left) to negative (right) → unstable equilibrium

At x_2: Force changes from negative (left) to positive (right) → stable equilibrium

FINAL ANSWER Option 2

CHEMISTRY

46)

$XeO_3 = sp^3$

$XeO_2F_2 = sp^3d$

47)

Question Explanation : Identify species with pπ-dπ bonding

Concept: pπ-dπ back bonding concept

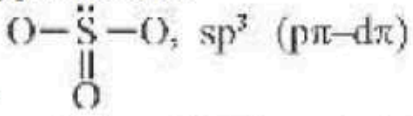
Solution:

• pπ-dπ bonding occurs when filled p orbitals of O overlap with vacant d orbitals of the central

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atom.

• In SO_3^{2-}, sulfur (3rd period) has vacant 3d orbitals that can overlap with O(2p) orbitals, forming pπ-dπ bonds.

$SO_3^{-2} \Rightarrow$ 

• In NO_3^-, BO_3^{3-}, and CO_3^{2-}, central atoms (N, B, C) belong to the 2nd period and do not have d orbitals available, so pπ-dπ bonding is not possible.

Final Answer: Option (2)

48) **Question Explanation:** Determining hybridization changes during an acid-base reaction.

Concept: Lewis Acid-Base Interaction and Hybridization

$$\underset{sp^2}{BF_3} + \underset{sp^3}{\ddot{N}H_3} \rightarrow \underset{sp^3}{F_3B} \leftarrow \underset{sp^3}{NH_3}$$

Solution:

In the reaction, NH_3 (donor) uses its lone pair to form a dative bond with BF_3 (acceptor).

Nitrogen remains sp^3 hybridized, but Boron changes its state.

BF_3 starts as sp^2 (trigonal planar) and becomes sp^3 (tetrahedral) in the adduct $[F_3B \leftarrow NH_3]$.

NH_3 is sp^3 before and after the reaction.

Final Answer: Hybridisation of only B changes

49) More than 8 valence electrons on central atom.

50) **Question Explanation:**

Identifying para-substituted benzene derivatives with non-zero dipole moments.

Concept:

Molecular Polarity and Dipole Moment (μ).

Solution:

For a molecule to have a net dipole moment $(\mu \neq 0)$, the individual bond dipoles must not cancel out.

In (a) p-dichlorobenzene and (b) p-dicyanobenzene, the substituents (–Cl and –CN) are linear or single-atom attachments.

The bond dipoles are exactly opposite (180°) and cancel out, so $\mu = 0$.

In (c) Hydroquinone and (d) p-dithiobenzene, the groups (–OH and –SH) are angular due to lone pairs on O and S.

Because these groups can rotate, they do not always align perfectly opposite, preventing the dipoles from canceling out completely. Thus, $\mu \neq 0$.

$\mu_{(a)} = 0$ (Linear $C-Cl$ vectors cancel)

$\mu_{(b)} = 0$ (Linear $C-C \equiv N$ vectors cancel)

$\mu_{(c)} \neq 0$ (Angular $-OH$ group rotation prevents cancellation)

$\mu_{(d)} \neq 0$ (Angular $-SH$ group rotation prevents cancellation)

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Final Answer: Option(3).

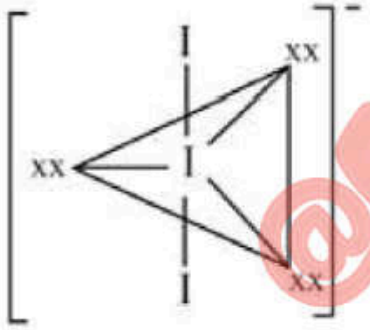
51) $$\text{Ionic} = \frac{\mu_{obs}}{\mu_{theo}} \times 100$$
$$= \frac{1.2 \times 10^{-18}\,\text{esu} \times \text{cm} \times 100}{4.8 \times 10^{-10} \times \text{esu} \times 10^{-8}\,\text{cm}} = 25\%$$
% of covalent = 100 - 25 = 75%

52)

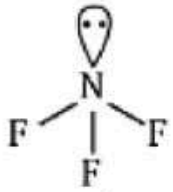
Compound	Boiling point (K)
H_2O	373
H_2S	213
H_2Se	232
H_2Te	269

- The boiling points of these Hydride,not exactly increases with increase in molar mass.
- H_2O has maximum boiling point due to intermolecular hydrogen bonding.

53) The geometric arrangement of orbitals is trigonal bipyramidal and the shape is linear.

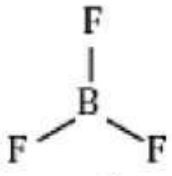


54)



sp^3

Trigonal pyramidal polar

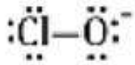


sp^2

Trigonal planar non-polar

55) B.L. $\mu \frac{1}{\text{B.O.}}$

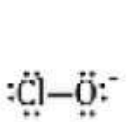
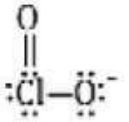
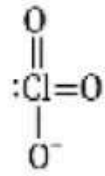
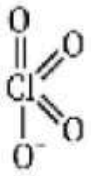
ClO^- ClO_2^- ClO_3^- ClO_4^-



$$\text{B.O.} = \frac{\text{Total no. of } (\sigma + \pi)}{\text{Total no. of Surrounding atom}}$$

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Hence

$B.O. = \frac{1}{1} = 1$ $\quad \frac{2+1}{2} = 1.5$ $\quad \frac{3+2}{3} = 1.66$ $\quad \frac{4+3}{4} = \frac{7}{4} = 1.75$

B.L. order $ClO^- > ClO_2^- > ClO_3^- < ClO_4^-$.

56)

Question Explanation:
Identify the factor that **raises the boiling point** of a compound.

Concept:
Stronger **intermolecular forces** lead to a **higher boiling point**.

Solution:

A. **Intermolecular hydrogen bonding**

A. Increases the attraction between molecules.
B. More energy is required to separate the molecules.
C. Therefore, the **boiling point increases**.

B. **High volatility**

A. Volatile substances evaporate easily.
B. They generally have **low boiling points**.

C. **Intramolecular hydrogen bonding**

A. Occurs within the same molecule.
B. Reduces intermolecular attraction.
C. Usually **lowers the boiling point**.

D. **Non-polarity**

A. Non-polar compounds have weak van der Waals forces.
B. Hence, they generally have **lower boiling points**.

Final Answer: Option (1)

57) XeF_4 has square planar shape. (5 atom in a plane)

58) **Explain Question:**
We are asked to identify which compound contains both ionic and covalent bonds.

Concept:
Ionic and covalent bonds.

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Solution:
• $C_6H_5Cl \rightarrow$ only covalent bonds.
• $H_2O \rightarrow$ only covalent bonds.
• $NaOH \rightarrow Na^+ - OH^-$ (ionic) and O–H (covalent.)
• $CO_2 \rightarrow$ only covalent bonds.

Final Answer: Option (3)

59)

Question Explanation : Identify species with sp^3d and one lone pair.

Concept : VSEPR theory

Solution :
(1) SF_4: S has 6 valence e^- , 4 bonds with F → 1 lone pair . Steric number = 5 → sp^3d see- saw. Correct
(2) $[PCl_4]^+$: P has 5 Valence e^- positive charge removes $1e^-$ → $4e^-$ all used in bond → no lone pair. Steric number = 4 → sp^3, tetrahedral . Incorrect
(3) XeO_2F_2 : Xe has 8 valence e^-, bonded to 2 O (double bonds) + 2F → $6e^-$ use (4 sigma bond) , leaving 1 lone pair . Steric number = 5 → sp^3d, see - saw , Correct
(4) $ClOF_3$: Cl has 7 valence e^-, bonded to 3F + 1 O (double bond) → $5e^-$ use (4 sigma bond), leaving 1 lone pair , Steric number = 5 → sp^3 d, see- saw . Correct

Final Answer : Option (2)

60)

Explanation -
The question asks which pair of molecular orbitals from the given options have the same energy level (are degenerate).

Concept - (Degenrate energy level)
π2px and π2py molecular orbitals are degenerate, meaning they have the same energy. They are formed from the sideways overlap of 2p atomic orbitals. Other orbital combinations listed have different energy levels.

Final Answer - Option (3)

61)

fact

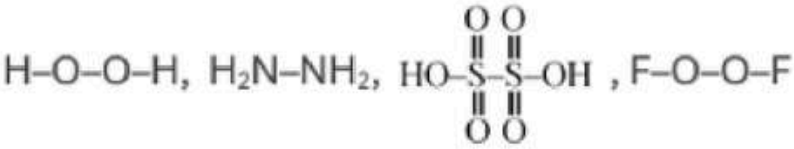
62)

Explain Question: Find molecules having both polar and non-polar bonds.

Concept: Bond polarity and electronegativity difference

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Solution:
H_2O_2, N_2H_4, $H_2S_2O_6$ and O_2F_2: contain nonpolar (N–N / S–S/ O–O) and polar (O–H / N–H / S–O/ O–F) bonds.



Final Answer: Option (4)

63) **Explaining :** Bond angle
Concept : This question is based on bond angle
Solution : a < c < b steric repulsion
Final Answer : (1)

64)

Explain Question: Asking about paramagnetic species.

Concept: MOT.

Solution:
(a) $N_2 \rightarrow$ Diamagnetic
(b) CO → Diamagnetic
(c) $B_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 (\pi 2p_x^1 = \pi 2p_y^1)$, (10e⁻s) unpaired e⁻s = 2 (Paramagnetic)
(d) $NO_2 \rightarrow$ O=N→O, NO_2 is a odd electron species. so it is Paramagnetic in nature.
both c and d

Final answer: option (3)

65) **Explaining :** isostructural
Concept : This question is based on shape
Solution : XeF_2, IF_2^- $2\sigma + 3\square p$ linear
$\Rightarrow$ 5 sp^3d
Final Answer : (1)

66) **Question Explanation:** Identify incorrect statements regarding symmetry in MOT.

Concept: Symmetry of Molecular Orbitals.

Solution: Statement (III) is incorrect because while **sigma (σ)** molecular orbitals are symmetrical around the bond axis (cylindrical symmetry), **pi (π)** molecular orbitals are not symmetrical around the bond axis; they have a nodal plane passing through the internuclear axis.

Final Answer: Option (4)

@EXAMLIX

67) **Question Explanation:** Determine the hybridization of Nitrogen in various species.

Concept: Hybridization of Nitrogen Compounds.

Solution:

A. NH_3: 3 Bonds + 1 Lone Pair = Steric Number 4 → **sp^3**
B. NH_4^+: 4 Bonds + 0 Lone Pairs = Steric Number 4 → **sp^3**
C. NO_2^+: O = N^+ = O (Linear), 2 Bonds + 0 Lone Pairs = Steric Number 2 → **sp**

Final Answer: Option (4)

68)

Explain Question: Compare covalent character of lithium halides.

Concept: Polarisation.

Solution: I^- is larger and more polarizable than F^-, so LiI shows more covalent character; both A and R are true and R explains A.

Final Answer: Option (1)

69)

Lattice energy LiF > NaF > KF > RbF
ionic character NaI < NaBr < NaCl < NaF
Polarising power $K^+ < Ca^{+2} < Cd^{+2}$
Polarisability $F^- < O^{-2} < Cd^{-2}$

@EXAMLIX

70)

Explain Question :
The question asks you to determine the correct order of the "extent" of hydrogen bonding in three molecules: HF (hydrogen fluoride), H_2O (water), and H_2O_2 (hydrogen peroxide).

Concept :
Extent of H-bonding depends on number of hydrogen atoms bonded to highly electronegative atoms and lone pairs available.

Solution :
The extent of H-bonding refers to the total number of hydrogen bonds a molecule can form H_2O_2 has two -OH groups, allowing for more H-bonds than H_2O. HF forms the fewest H-bonds due to its single hydrogen. Therefore, the correct order is $H_2O_2 > H_2O > HF$.

Final Answer : Option (2)

71) **Explain Question:** Identify compound with highest lattice energy among given salts.

Concept: This question is based on Lattice energy.
Solution: Lattice energy increases with higher ionic charge and smaller ionic radii. Al^{3+} and F^- have
highest charge product and smallest size, so AlF_3 has maximum lattice energy.

$$\text{Lattice energy} \propto \frac{|z^+.z^-|}{r^+ + r^-}$$

Final Answer: 1

72) **Explanation Question:**
We are asked about the solid-state structure of phosphorus pentachloride (PCl_5).

Concept:
Ionic structure in solid state

Solution:
- In solid state, PCl_5 does not exits as discrete molecules.
- It ionizes to from PCl_4^+ (tetrahedral) and PCl_6^- (octahedral) ions.
- This stabilizes the lattice via ionic interactions.

Final Answer: Option (3)

73)

Explain Question: Identify which pair of compounds is incorrectly matched with its property.
Concept: This question is based on Hydrogen bonding and molecular interactions
Solution:
o-Nitrophenol forms **intramolecular H-bond** → higher vapour pressure, lower boiling point (1 & 2 correct).
Glycerol has higher viscosity (3 correct).
But **H_2O > HF** in boiling point due to more H-bonds per molecule → so (4) is incorrect.
Final Answer: Option (4)

74)

Question explanation : We need to identify which interaction represents the strongest attraction between molecules or ions.
Concept :
Ion-dipole interactions are stronger than hydrogen bonding and all types of van der Waals forces.
Solution
H_2—H_2: Nonpolar molecules → weak London dispersion forces.
H-Cl—C_6H_6: Polar + nonpolar → dipole-induced dipole → weak.
H-Cl—H-Cl: Dipole-dipole interaction → moderate strength.
Na^+—Cl-H: Ion-dipole interaction → strongest among all listed.
Final answer Option 4 (Na^+ — Cl-H)

75)

@EXAMLIX

Question Explanation: Arrange chlorides according to their covalent character.

Concept Used: Fajan's Rule

Answer Explanation: Covalent character increases with smaller cation size and higher polarising power. Among alkaline earth metals, cation size increases down the group, so covalent character decreases: **$BeCl_2$ > $MgCl_2$ > $CaCl_2$ > $SrCl_2$**.

Final Answer: Option (3)

76) **Explain Question:** Which of the following has both covalent and coordinate bonds?

Concept: This question is based on dative (coordinate) bond.

Solution:
(a) $N_2H_5^+$ → has N–N covalent and N → H^+ **coordinate bond.**
(b) H_2CO_3 → only covalent bonds.
(c) N_2O_5 → N–O covalent and N → O **coordinate bond.**
(d) H_3O^+ → two O–H covalent, one O → H^+ **coordinate bond.**
(e) SiF_6^{2-} → Two **coordinate bond.**

Final Answer: Option (2)

@EXAMLIX

77)

Explain Question : Count total antibonding electrons for O_2^-, O_2, and O_2^{2-} respectively.
Concept : This question is based on Molecular Orbital Theory
Solution :
O_2 ($16e^-$) has 6 antibonding electrons (ABMO). O_2^- ($17e^-$) adds one electron to an ABMO, for a total of 7. O_2^{2-} ($18e^-$) adds two, for a total of 8.
O_2 : (16 electron)- $\sigma1s^2$, σ^*1s^2, $\sigma2s^2$, σ^*2s^2, $\sigma2pz^2$, $\pi2px^2 = \pi2py^2$, $\pi^*2px^1 = \pi^*2py^1$
O_2^- : (17 electron)- $\sigma1s^2$, σ^*1s^2, $\sigma2s^2$, σ^*2s^2, $\sigma2pz^2$, $\pi2px^2 = \pi2py^2$, $\pi^*2px^2 = \pi^*2py^1$
O_2^{2-} : (18 electron)- $\sigma1s^2$, σ^*1s^2, $\sigma2s^2$, σ^*2s^2, $\sigma2pz^2$, $\pi2px^2 = \pi2py^2$, $\pi^*2px^2 = \pi^*2py^2$
Final answer : 1

78)

Explain Question:
Asking about spices having only π bond.

Concept:
MOT

Solution:
According to M.O.T. electronic configuration of C_2 molecule is -
$\sigma1s^2 < \sigma^*1s^2 < \sigma2s^2 < \sigma^*2s^2 < \pi_2p_x^2 = \pi_2 p_y^2$
so, C_2 molecule contain only 'π' bond

Final answer: option (3)

79) **Question Explanation:**
Find the increasing order of s-character in hybrid orbitals by determining the hybridization of each given molecule or ion.

Concept: Hybridization determines percentage s-character.

Solution:
The percentage s-character is calculated using the formula:

$$\% \text{ s-character} = \frac{1}{\text{Total number of hybrid orbitals}} \times 100$$

1. (II) XeF_4 : Steric number = 4 bonds +2 lone pairs $= 6\left(sp^3d^2\right)$. % s-character $= \frac{1}{6} \times 100 \approx 16.67\%$

2. (III) I_3^- : Steric number = 2 bonds +3 lone pairs $= 5\left(sp^3d\right)$. % s-character $= \frac{1}{5} \times 100 = 20\%$

3. (IV) NCl_3 : Steric number = 3 bonds +1 lone pair $= 4\left(sp^3\right)$. % s-character $= \frac{1}{4} \times 100 = 25\%$

4. (I) CO_3^{2-} : Steric number = 3 sigma bonds +0 lone pairs $= 3\left(sp^2\right)$. % s-character $= \frac{1}{3} \times 100 \approx 33.33\%$

5. (V) $BeCl_2(g)$: Steric number = 2 bonds +0 lone pairs = 2(sp). % s-character $= \frac{1}{2} \times 100 = 50\%$

Increasing order: II < III < IV < I < V

Final Answer: Option(1)

@EXAMLIX

80)

Explanation Question:
We are asked about the role of formal charge in deciding the most stable (lowest energy) Lewis structure.
Concept:
This question is based on Formal charge (FC)
Solution:
Formal charge (FC) is a hypothetical charge assigned to atoms in a Lewis structure.
The most stable structure usually has minimum formal charges on atoms and a proper distribution of charges. FC helps compare possible structures and pick the lowest energy one.
Statement-I: Correct, because FC indeed helps in selecting the lowest energy structure from many possibilities.
Statement-II: Correct, because generally the most stable structure has the smallest (closest to zero) formal charges on atoms.
Both Statement-I and Statement-II are correct.
Final Answer: (2)

81)

In I, lp occupies equitorial position &

∴ have minimumrepulsion
∴ most stable
Option 1

82)

$$\text{Ionic conductance} \propto \frac{1}{\text{Extent of hydration}}$$

83)

Question Explanation: Evaluates the correctness of statements regarding the overall molecular polarity and individual bond polarities of CO_2 and SO_2.

Concept Used: Dipole Moment & Bond Polarity

Answer Explanation:

A. **Statement 1:** CO_2 is a non-polar molecule due to its linear geometry (net dipole moment = 0 D), whereas SO_2 is polar due to its bent geometry (net dipole moment ≠ 0 D). Thus, S1 is incorrect.

B. **Statement 2:** Both C=O and S=O bonds involve atoms with differing electronegativities, meaning both molecules contain polar bonds. Thus, S2 is correct.

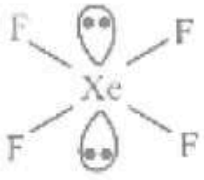
Final Answer: Option (4)

@EXAMLIX

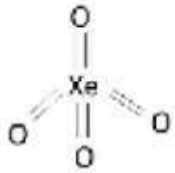
84) **Explain Question:**
In this question choose property which is not change after changing XeF_4 into XeO_4.

Concept:
Hybridisation, Dipole moment, oxidation state.

Solution:



Sp^3d^2 planar xe = +4
Dipole = 0 Moment



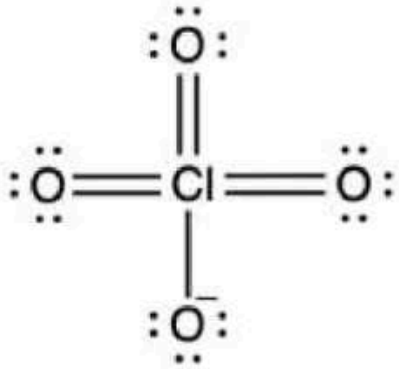
Sp^3 non–planar xe = +8
Dipole = 0 Moment

Final Answer: Option (4)

85)

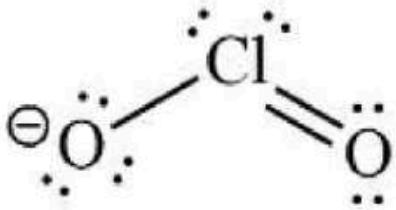
Explain the question : Asking about compounds having the same hybridisation.
Concept : This question is based on hybridisation

Solution :



ClO_4^-

sp^3 (4σ + 0 lp)



ClO_2^-

sp^3 (2σ + 2 lp)

Final answer : Option (1)

86)

Question Explanation:
Order of bond strength based on orbital overlapping.

Concept:
VBT and Overlapping

Solution:
Bond strength depends on overlapping extent. Smaller size gives better overlap. If size difference is similar, directional nature decides overlap. Since p-orbital has better directional character than s-orbital:

1s–1s > 1s–2p > 1s–2s

Final Answer: Option (4)

@EXAMLIX

87)

For formation of Ionic compound IE of A should be less and EA of B should be maximum.

88)

Dipole moment (D.M.) of H_2O is not equal to zero.
Hybridization :
H_2O SN = δ + l.p. = 2 + 2 = 4



[D.M. ≠ O]

89)

Question Explanation: Identifies which pair of species has unpaired electrons residing in

antibonding molecular orbitals.

Concept Used: Molecular Orbital Theory

Answer Explanation:

A. **O_2** (16 electrons): Electronic configuration is $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2pz^2\ (\pi 2px^2 = \pi 2py^2)\ (\pi^* 2px^1 = \pi^* 2py^1)$. Unpaired electrons are in the antibonding π^* orbitals.

B. **He_2^+** (3 electrons): Electronic configuration is $\sigma 1s^2\ \sigma^* 1s^1$. *The single unpaired electron is in the antibonding* $\sigma^* 1s$ orbital.

C. **C_2** (12 electrons): Has no unpaired electrons (diamagnetic).

D. **N_2** (14 electrons): Has no unpaired electrons (diamagnetic).

Final Answer: Option (2)

90)

SF_4 → see saw shape
CCl_4 tetrahedral; no l.p. at central atom & all terminal atoms are same, so all bond angles are same.
$CHCl_3$, hybridisation sp^3 but all terminal atoms are not same.
XeF_6 hybridisation sp^3d^3 (all bond angles are not same) – Capped octahedral.

@EXAMLIX

BOTANY

91) **Question Explanation:** Question is asking: Which structure is not formed during sexual reproduction in fungi.

Concept: Reproduction in fungi (General characters of fungi).

Solution: Zoospore is asexual, not sexual.

Final answer: Option (2)

92) NCERT Pg. # 20, fig. 2.3

93)

Question Explanation: Question asks about Identifying the figures (a, b, c) with their correct features (i-iii).

Concept:

Solution: (a) → Mucor(Zygomycetes) → mycelium is aseptate → matches (i).

(b) → Aspergillus (Ascomycetes) → sexual spores (ascospores) are produced endogenously inside asci → matches (iii).

(c) → Agaricus (Basidiomycetes) → asexual spores are generally not found → matches (ii).

Final answer: 2

94)

Question Explanation: Question asking about identifying bacteria according to their shape from the given figures.

Concept: Morphology of Bacteria

Solution:

Figure A → comma shaped → Vibrio

Figure B → rod shaped → Bacilli

Figure C → spherical → Cocci

Figure D → spiral → Spirilla

Correct matching: A = Vibrio, B = Bacilli, C = Cocci, D = Spirilla → matches Option 4.

@EXAMLIX

Final answer: Option 4

95)

NCERT pg.no. 15, 17 and 18

96) **Explain Question:** The question asks which of the given fungus–group matches is correct.

Concept: Fungal classes

Solution: Alternaria → Deuteromycetes (no sexual reproduction) -correct

Mucor → Zygomycetes (not Ascomycetes) - incorrect

Agaricus → Basidiomycetes, but it is saprophytic, not parasitic -incorrect
Phytophthora → Oomycetes, not Basidiomycetes -incorrect

Final answer: (1)

97)

Question explanation: The question asks to identify A, B, and C in the classification table of Monera, Protista, Fungi, and Plantae.

Concept: Five kingdom system

Solution: A represents the fungal cell wall → Chitin

B represents the nuclear membrane in Monera → Absent

C represents body organisation in fungi → Loose tissue

Final answer: 3

98)

Question Explanation: Question asks which feature is NOT used by R.H. Whittaker as a basis for the five-kingdom classification system

Concept: Five Kingdom classification

Solution: Whittaker's used 5 criteria:
1. Cell type
2. Cell organization
3. Body organization
4. Mode of nutrition
5. Reproduction
He did not use reserve food materials as a criterion.

@EXAMLIX

Final Answer: option (3)

99)

Question explanation: The question asks which option correctly identifies the labelled parts (A, B, C, D) of a bacteriophage diagram.

Concept: Virus

Solution: According to the given diagram, the correct identification sequence is:

A – Head , B – Sheath ,C – Collar , D – Tail fibres

Final answer: 1

100)

Question explanation: The question asks whether the reason correctly explains why the two-kingdom system was inadequate.

Concept: Two-kingdom classification

Solution: Two-kingdom system failed because it did not separate:- Prokaryotes vs. eukaryotes

,

Unicellular vs. multicellular organisms ,Green algae (autotrophs) vs. fungi (heterotrophs)

Both Assertion and Reason are true, and the Reason correctly explains the Assertion.

Final answer: Both A and R are true and R is the correct explanation of A.

101)

Question explanation: The question asks which of the given statements is incorrect about classification systems.

Concept: History of taxonomy

Solution: Statement 4 is incorrect because reproduction and phylogeny were also important criteria in Whittaker's system.

Final answer: The main criteria of five kingdom classification is cell structure, body organisation, mode of nutrition but not Reproduction and Phylogenetic relationship.

102)

NCERT Pg. # 22 to 24

103) NCERT-XI, Pg. # 24

104) NCERT (XIth) Pg. # 23, 1st para

105) NCERT-XIth Pg. # 24

106)

NCERT-XIth Pg # 20

107) NCERT-XIth Pg # 19

108) NCERT (XIth) Pg. # 23

109) NCERT XIth Pg.#26

110) NCERT XIth Pg.#21

@EXAMLIX

111) NCERT Pg. # 2.3/22

112) NCERT-XI, Pg. # 15

113) NCERT XIth Pg. # 23

114)

NCERT XI Pg # 27

115) NCERT XI Pg.# 19, 20

116)

The kingdom of unicellular eukaryotes is Protista, while the kingdom of unicellular prokaryotes is Monera.

The correct answer is:OPTION(3)Protista and monera

117) NCERT XI, Pg. # 11, 12

118) **Explain Question:**
The question asks which prokaryote lacks a cell wall around the cell membrane.
Concept:
This question is based on Mycoplasma
Explanation:
Mycoplasma → A group of prokaryotes (bacteria-like) that lack a cell wall. Their plasma membrane has sterols for protection instead.
Final Answer: (4)

119)

Question Explanation: Question asks whether each statement about the three-domain system and Whittaker's five-kingdom system is correct.

Concept: Three-domain system

Solution:Statement-1 :- The three-domain system divides the Kingdom Monera into three domains-false because Monera splits into two domains: Archaea and Bacteria. Statement. 2. Five-kingdom system was proposed by R.H. Whittaker (1969)-true.

Final answer: 2 (Only statement-2 is correct)

120)

@EXAMLIX

Explain Question:
The question asks whether the assertion and reason about Deuteromycetes (imperfect fungi) are correct and related.

Concept:
Deuteromycetes.

Solution:
Assertion is true → They are called imperfect fungi because their sexual stage is not known.
Reason is false → The statement is incorrect; it should be "sexual stage is absent/unknown," not "only sexual or vegetative phases are known".

Final answer: Option (3)

121) **Question Explanation:** Question asks which statements about slime moulds (their nutrition and structures formed during unfavourable conditions) are correct.

Concept: Slime moulds

Solution: Statement 1-Slime moulds are saprophytic protists → TRUE.
Statement 1I-Plasmodium is formed during favourable conditions (vegetative stage), not during unfavourable ones-FALSE

Final answer: 1 (Only statement 1 is correct)

@EXAMLIX

122)

Explain Question:
The question asks to correctly match organisms/features in Column I with their descriptions in Column II.

Concept:
Kingdom Protista.

Solution:
Chief producer of ocean → Diatoms (IV), as they form the major part of phytoplankton.
Red tides → Gonyaulax (III), a dinoflagellate responsible for red tide formation.
Pellicle layer → Euglena (II), which has a protein-rich pellicle instead of a cell wall.
Plasmodium → Slime mould (I), where the vegetative phase is a plasmodium.

So the correct matching is: A-IV, B-III, C-II, D-I.

Final answer: Option (3)

123) NCERT Pg. # 27

124)

Explain Question:
The question asks why cyanobacteria are used in agricultural fields for crop improvement.

Concept:
Cyanobacteria.

Solution:
Cyanobacteria (like Anabaena, Nostoc) are photosynthetic prokaryotes that can fix atmospheric nitrogen into ammonia, enriching soil fertility. They are used as biofertilizers in paddy fields. Respiration and algal blooms are not the reasons for crop improvement; the main benefit is N_2 fixation.

Final answer: Option (3)

125)

Question Explanation: Question asks about how many kingdoms contain prokaryotic organisms in Carl Woese's classification.

Concept: Carl Woese's Classification

Solution: Woese's separated prokaryotes into two groups (Archaebacteria & Eubacteria), prokaryotes occur in two kingdoms.

Final Answer: option (1)

126) **Explain Question:** Question asking select correct statement about lichen.
Concept: This question is based on Lichen.
Solution: (A) Lichens are mutually useful associations between algae and fungi.**-Correct**

A. Lichens are a symbiotic relationship between algae and fungi.

(B) Algae prepares food for fungi, and the fungi provide shelter to its partner. **Correct**

A. The algal partner (phycobiont) performs photosynthesis, while the fungal partner (mycobiont) provides protection and absorbs water and minerals.

(C) Lichens are very good pollution indicators. **Correct**

A. Lichens are highly sensitive to air pollution, Lichens are not found in polluted areas.
B. Excellent bioindicators of air quality.

(D) The alga is phycobiont, and the fungus is mycobiont. **Correct**

A. Phycobiont = Algal component (performs photosynthesis).
B. Mycobiont = Fungal component (provides structure and protection).

Final answer : 4

127)

Explain Question: Question asks about the feature that makes archaebacteria different from other bacteria.

@EXAMLIX

Concept: Archaebacteria.

Solution: Their major difference is in cell wall and membrane structure, not in cellular organization or parasitic nature.

Final Answer: Option (1)

128)

Solution/Explanation :

Morels and **truffles** are not poisonous, in fact, they are considered **delicacies** and are highly prized in cooking. However, it's important to note that some species of morels **must be cooked** properly, as raw morels can be toxic to humans. Despite this, they **are not inherently poisonous** when cooked, and they are not toxic like certain other types of mushrooms eg. Amanita species.

129) NCERT Pg. # 18

130) **Question Explanation:** Question asks which statement correctly describes viruses.

Concept: Virus

Solution: 1. Free DNA/RNA without protein coat → Viroids, not viruses.
(F) 2. Abnormally folded proteins → Prions, not viruses.(F)
3. Their genetic material is always enclosed within a protein coat (capsid).(F)
4.Viruses always contain either DNA or RNA, never both.(T)

Final answer: 4 (They have DNA or RNA with protein coat.)

@EXAMLIX

131) **Question Explanation:** Identify which kingdom has heterotrophic, eukaryotic, multicellular organisms without cell wall.
Concept: Five kingdom system.
Solution:

A. **Fungi** → Heterotrophic, Eukaryotic, and Multicellular, but have cell wall (chitin)
B. **Plantae** → Autotrophic with cell wall
C. **Animalia** → Heterotrophic, Eukaryotic, and Multicellular, no cell wall

Only Animalia fits all criteria.

Final Answer: option (3)

132)

Question Explanation: In which year did R.H. Whittaker proposed the five-kingdom classification?

Concept: Five-kingdom system

Solution: Whittaker's well-known five-kingdom system (Monera, Protista, Fungi, Plantae, and Animalia) was proposed in 1969.

Final Answer: option (3)

133)

As per NCERT Page number:- 13,14

134) As per NCERT Page number:- 13

135)

Question Explanation :
The question asks which feature is exclusive to the Monera kingdom and not shared with any other kingdom in R.H. Whittaker's five-kingdom classification.

Concept :
R.H. Whittaker's five-kingdom classification.

Solution :
Only nitrogen fixation is a unique functional trait primarily seen in certain prokaryotes (Monera).

Final Answer :
Option (4)

@EXAMLIX

ZOOLOGY

136) **Question Explanation:** What will happen if the stretch receptor of urinary bladder are totally removed.
Concept : Micturition.
Explanation: Stretch receptors in the bladder wall play a crucial role in initiating the micturition reflex. When the bladder fills with urine, the walls stretch, activating these receptors. This sends signals to the spinal cord and brain, triggering the urge to urinate. If these stretch receptors are removed, the brain will not receive the signal that the bladder is full, even if it is. As a result, the micturition reflex will not be initiated, and the individual will not experience the urge to urinate. Therefore, without the stretch receptors, there will be no micturition.
The correct answer is : There will be no micturition.

137) **Solution :** Podocytes form the visceral/inner layer of Bowman's capsule and leave filtration slits. The glomerulus is formed from branches of the afferent arteriole, while blood exits through the efferent arteriole.
Final Answer : (3)

138)

Ncert Page 290.

139)

Ncert Page 209.

140) **Explain Question:** Excretory product of aquatic amphibians and reason.
Concept: Introduction And Types Of Excretory Matter
Solution:
Aquatic amphibians (e.g., tadpoles) are ammonotelic, excreting ammonia.
Ammonia is highly soluble but requires a large volume of water for safe elimination, which is possible in aquatic habitat.Both Assertion and Reason are correct, and Reason explains Assertion.
Final answer: Option 1: Both (A) and (R) correct, and (R) explains (A)

141)

NCERT - Pg. # 293

142) Kidney stones are produced due to crystallisation of oxalates.

143)

Question explanation: Identify the site from which urea moves into the medullary interstitium to maintain osmolarity.
Concept: Excretion – Urine concentration

Solution:

A. **Urea recycling** helps **maintain the osmotic gradient** in the renal medulla.

B. Small amounts of urea diffuse from the **distal part of the collecting duct** into the **medullary interstitium**.

C. Other segments (PCT, thin/thick loop) do not significantly contribute to urea movement.

Final answer: Option 4

144) NCERT (XIth) Pg. # 296

145)

@EXAMLIX

List-1		List-2	
(A)	Glycosuria	(ii)	Presence of glucose in urine
(B)	Ketonuria	(iv)	Presence of ketone bodies of urine
(C)	Glomerulonephritis	(i)	Inflammation of glomeruli of kidney
(D)	Haematuria	(iii)	Presence of blood in urine

146)

NCERT Page # 293

147) NCERT pg no.296

148) NCERT Pg. # 299

149) **Solution :** Less water reabsorption from collecting duct = More dilute urine will form.
Final Answer : (2)

@EXAMLIX

150)

As per NCERT page number:- **241**

151)

Question Explanation: Mammals and urine concentration.
Concept: Counter-current mechanism
Solution: Mammals can produce concentrated urine (Assertion is correct). This is because the counter-current mechanism involving the loop of Henle and vasa recta helps to maintain a concentration gradient in the medullary interstitium (Reason is correct). The reason provides the correct explanation for the assertion.
Final Answer: Assertion is correct, reason is correct & reason is correct explanation of assertion.

152) **Explanation:** The process of urination (micturition) follows these steps in order:
Bladder fills with urine and becomes distended (Step II)
Stretch receptors in the bladder wall send signals to CNS (Step I)
CNS sends motor signals to contract bladder muscles and relax urethral sphincter (Step IV)
Urination (micturition) occurs (Step Ill)
Thus, Option 3 (II → I → IV → Ill) is the correct answer.

153)

The questionlasks for the anatomical name of the cortical tissue that extends into the medullary region between the renal pyramids.
Solution AND CORRECT ANSWER:

Columns of Bertini (Renal columns): These are inward extensions of the renal cortex that separate the renal pyramids.
Therefore, the correct answer is Columns of Bertini.
Correct Answer: 4. Columns of Bertini
Difficulty Level: Easy

154)

Explanation - Asking when JG cells get activated: fall in GFR, low glomerular BP, or high renal artery BP.

Concept - RAAS

Solution -

(A)Fall in GFR → JG cells activated.
(B)Decrease in glomerular BP → JG cells activated.
(C)Increase in renal artery BP → JG cells not activated.
Final Answer - A & B

155) Answer (3)
Convoluted tubules : Proximal convoluted tubules (PCT) and distal convoluted tubules (DCT) are present in cortex while rest are in medulla which comprises renal pyramids.

156)

Question Explanation -The question asks for the correct sequence of steps in haemodialysis.

Concept - Process of haemodialysis

Solution -

A. Blood is drained from an artery and anticoagulant (heparin) is added → I

B. Blood passes through coiled porous cellophane membrane in dialysis fluid → III

C. Nitrogenous wastes are removed from blood → II

D. Blood is mixed with antiheparin and returned to vein → IV

Correct order: I → III → II → IV

Final Answer - Option 3

157)

@EXAMLIX

Question Explanation: This question is asking about the source and stimulus for ANF secretion.

Concept: Hormonal regulation of kidney function

Solution: ANF is secreted by the atrial wall of the heart when venous return (blood volume) is high, leading to increased sodium and water excretion to reduce blood volume and blood pressure.

Final answer: 1. Atrial wall when the venous return is high

158)

Question explanation
The question is asking to match different animals with their correct excretory organs.

Concept: Excretory organs in animals.

Solution
(i) Earthworm → (a) Nephridia
(ii) Cockroach → (c) Malpighian tubules
(iii) Prawn → (d) Green glands
(iv) Amphioxus → (b) Protonephridia

Final answer: Option 1

@EXAMLIX

159) NCERT-XI, Pg. # 212

160)

Explain Question : This question is asking about substances that are actively reabsorbed in the nephron.

Concept : This question is based on Reabsorption in Nephron.

Solution : Glucose is actively reabsorbed in the proximal convoluted tubule. Chloride ions, potassium, and nitrogenous wastes are reabsorbed passively, not actively.

Final Answer : option (2) Only a

161) **Explanation:** Each kidney has nearly one million nephrons.
Final Answer: (4)

162)

Question explanation:
The question is asking to correctly match parts of the nephron in Column-I with their

corresponding functions in Column-II.

Concept: mechanism of urine formation.

Solution:
A – PCT reabsorbs 70–80% of electrolytes and water → ii
B – DCT is involved in selective secretion of H+ and K+ ions → i
C – Loop of Henle helps concentrate urine → iv
D – Ascending limb is permeable to salts, not water → v
E – Collecting duct allows passage of small amount of urea → iii

So the correct match is: A – ii, B – i, C – iv, D – v, E – iii
Final answer: 3

163)

Explain Question : In the collecting duct, which ions are selectively secreted?

Concept : This question is based on mechanism of urine formation.

Solution : H^+ and K^+ are selectively secreted to regulate pH and potassium levels.

Final Answer : option (2) H^+ & K^+

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164)

NCERT PAGE NO. 210

165)

(a), (b) and (c) only
NCERT XI Pg No - 212

166)

Question explanation : The question is asking which substances are involved in the process of producing concentrated urine in the collecting duct.
Concept : Mechanism of urine formation..
Solution : In the collecting duct, a large amount of water is reabsorbed to make urine concentrated. It also allows a small amount of urea to pass into the medullary interstitium to maintain osmolarity.Final answer
4. A-reabsorbed, B-concentrated, C-urea

167)

NCERT Pg No # 210

168) As per NCERT Page number:- 213

169)

Explain Question : This question is asking about the segment of nephron where water reabsorption is minimum.

Concept : This question is based on Reabsorption in Nephron.

Solution :
The descending limb of Henle's loop is highly permeable to water, DCT and collecting duct allow conditional water reabsorption, but the ascending limb of Henle's loop is impermeable to water, hence shows minimum water reabsorption.

Final Answer : option (2) Ascending limb of loop of Henle

170) NCERT Pg. # 293 last para

171) NCERT (XI) Pg. # 292

172)

Question explanation: Identify the structures that form the juxta-glomerular apparatus (JGA).
Concept: Excretory system - Juxta-glomerular apparatus

Solution:

A. **Juxta-glomerular apparatus (JGA)** regulates **glomerular filtration rate (GFR) and blood pressure**.

B. It is formed by:

   A. **Modified smooth muscle cells of the afferent arteriole** → called **juxtaglomerular cells** (secrete renin)

   B. **Modified cells of the distal convoluted tubule (DCT)** → called **macula densa** (detect Na^+ concentration)

Final answer: Option 1

173) **Question Explanation :** Choose the correct statements type question.
Concept : Mechanism of urine formation and canter current.
Solution :
• Statement A - Correct: The Juxta-medullary nephron plays a crucial role in urine concentration due to its long Loop of Henle, which extends deep into the medulla, facilitating water reabsorption.

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• Statement B - Correct: The afferent arteriole has a larger diameter than the efferent arteriole, creating high glomerular pressure essential for filtration.
• Statement C - Incorrect: Renin (not Rennin) is an enzyme secreted by the Juxtaglomerular Apparatus (JGA) and is involved in the Renin-Angiotensin-Aldosterone System (RAAS), but it itself is not a vasoconstrictor-it catalyzes the conversion of angiotensinogen to angiotensin I, which is further converted to angiotensin II , a potent vasoconstrictor. Thus, Option 3 (A, B are correct while C is incorrect) is the right answer.
Final Answer : Option (3)

174)

Concept :
Integration of urea metabolism and clinical diagnosis.

Solution:
Urea is the main nitrogenous waste; its excretion is normal.
Urine analysis is used for metabolic disorders (like Diabetes).
Micturition is neurally controlled, not primarily endocrine.

Final Answer: Option (3)

175) NCERT-XI, Pg. # 297

176) **Question Explanation :**
Asks you to identify which of the given statements about hormones involved in blood pressure and fluid balance are correct.

Explanation : (A) Atrial natriuretic factor can cause lower in blood pressure: Atrial natriuretic factor (ANF) is released by the atria of the heart in response to increased blood volume and pressure. Its primary effects are to promote natriuresis (sodium excretion) and diuresis (water excretion), leading to a decrease in blood volume and consequently a lowering of blood pressure. Therefore, this statement is
(B) Angiotensin II is vasoconstrictor: Angiotensin II is a potent vasoconstrictor meaning it narrows blood vessels. This narrowing increases peripheral resistance leading to an increase in blood pressure.

Therefore, the statement is correct. **(C) ANF is a vasodilator:** ANF can also cause vasodilation, relaxing smooth muscles in blood vessels, which contributes to the lowering of blood pressure. Therefore, this statement is correct.
The option that includes all the correct statement is correct option 2.

177) **Question Explanation:** The question asks to complete the statement about the regulation of ADH (antidiuretic hormone) secretion when body fluid volume increases.

Concept: ADH Regulation

Solution:
When body fluid volume increases, the osmoreceptors in the hypothalamus are switched off because the osmotic concentration of body fluids decreases. As a result, the release of ADH is

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suppressed, reducing water reabsorption in the kidneys. This represents a negative feedback mechanism that helps maintain water balance in the body.

Therefore:

A. A = Osmoreceptors

B. B = Suppress

C. C = Feedback

Final answer: Option - 2

178) **Explanation:**
(i) Liver, the largest gland in our body secretes cholesterol, degrade steroid hormones, vitamins and drugs:

A. This statement is correct. The liver is the largest gland in the body and plays a major role in metabolism. It synthesizes cholesterol, and it is involved in the breakdown of steroid hormones, vitamins, and drugs.

(ii) Sebaceous glands eliminate certain substances like urea, sterols, and sebum:

A. This statement is incorrect. Sebaceous glands primarily secrete sebum (an oily substance) to lubricate the skin. They do not significantly excrete urea or sterols.

(iii) Sweat produced by sweat glands is an oily fluid containing NaCI:

A. This statement is incorrect. Sweat produced by sweat glands is mainly water with salts like sodium chloride (NaCl), and it is not oily. Sweat also helps in thermoregulation.

(iv) In uremic patients urea and glucose can be removed by a process called hemodialysis:

A. This statement is partially correct. Hemodialysis is a process used to remove waste products like urea and excess fluids from the blood in patients with kidney failure (uremia). However, glucose is not typically removed during hemodialysis unless specifically required in a certain medical context.

The correct answer is **Option 3: i only**.

179)

Question Explanation: The question asks to identify the correct statements about blood vessels associated with human nephrons.

Concept: The question is based on types of nephron

Solution: The correct answer is option 3.

A. Statement (a) is correct: Vasa recta runs parallel to the loop of Henle in juxtamedullary

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nephrons.

B. Statement (b) is correct: Cortical nephrons have no or highly reduced vasa recta.

C. Statement (c) is correct: Vasa recta, peritubular capillaries, and glomerulus all contain blood.

D. Statement (d) is incorrect: The blood vessel leading to the glomerulus is called the afferent arteriole, not efferent arteriole.

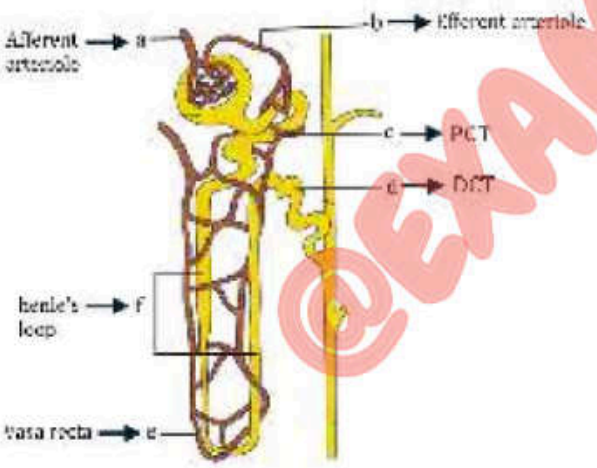
Final answer: Option 3

180)

Explain Question : Recognise the figure and find out the correct matching.

Concept : This question is based on Structure of nephron.

Solution :



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Final Answer : option (1) c-PCT, d-DCT, a-afferent arteriole, b-efferent arteriole, f-henle's loop, e-vasa recta